

DESIGN OF ADJUSTABLE DIGITAL FILTER FOR SDR APPLICATION

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OUTLINE

- Objective
- Introduction
- Block Diagram
- Simulation Waveform
- Performance Comparison
- Tools
- References

OBJECTIVE

- To implement various decimation approaches using
 - i) Coefficient Decimation Method (CDM),
 - ii) Modified Coefficient Decimation Method (MCDM),
 - iii) Improved Coefficient Decimation Method (ICDM).
- To implement Pipelined ICDM architecture to reduce delay.

INTRODUCTION

- Variable Digital filter is used for extraction of individual radio channels corresponding to multiple wireless communication.
- It is based on the improved coefficient decimation method(ICDM).
- A combination of CDM and MCDM is called ICDM.
- Mux I is based on CDM operation and Mux II is based on MCDM operation.
- The adder/subtractor block are used to perform sign reversal of appropriate coefficient in the MCDM.
- The decimation selector block is used to select signal of mux I and mux II.
- Pipelined ICDM uses to reduce time consumption.
- Decimation Selector block is used to select the ICDM I and ICDM II signal.

Contd...

- An input sine signal value is stored in LUT and coefficient value taken from a MATLAB tool. When the decimator factor $M=1$, then all the coefficient is multiplied with input signal.
- When the decimator factor $M=2$, then the odd coefficient is multiplied with input signal. After the multiplexer gets the value from multiplied output.
- When the decimator factor $M=3$, then the h_0, h_3, h_7 coefficient is multiplied with input signal. After the multiplexer gets the value from multiplied output.
- When the decimator factor $M=4$, then the h_3, h_7 coefficient is multiplied with input signal. After the multiplexer gets the value from multiplied output and so on.
- No of coefficient multiplication required to implement the decimated filter N/M . So, it reducing no of multiplication require to implement the filter.
- As M is increased there is decreased in complexity due to replacement of filter coefficient by zero.

Improved Coefficient Decimation Method (ICDM) :

- Mux I is based on ICDM I operation and Mux II is based on ICDM II operation.
- The decimation selector block is used to select signal of mux I and mux II.

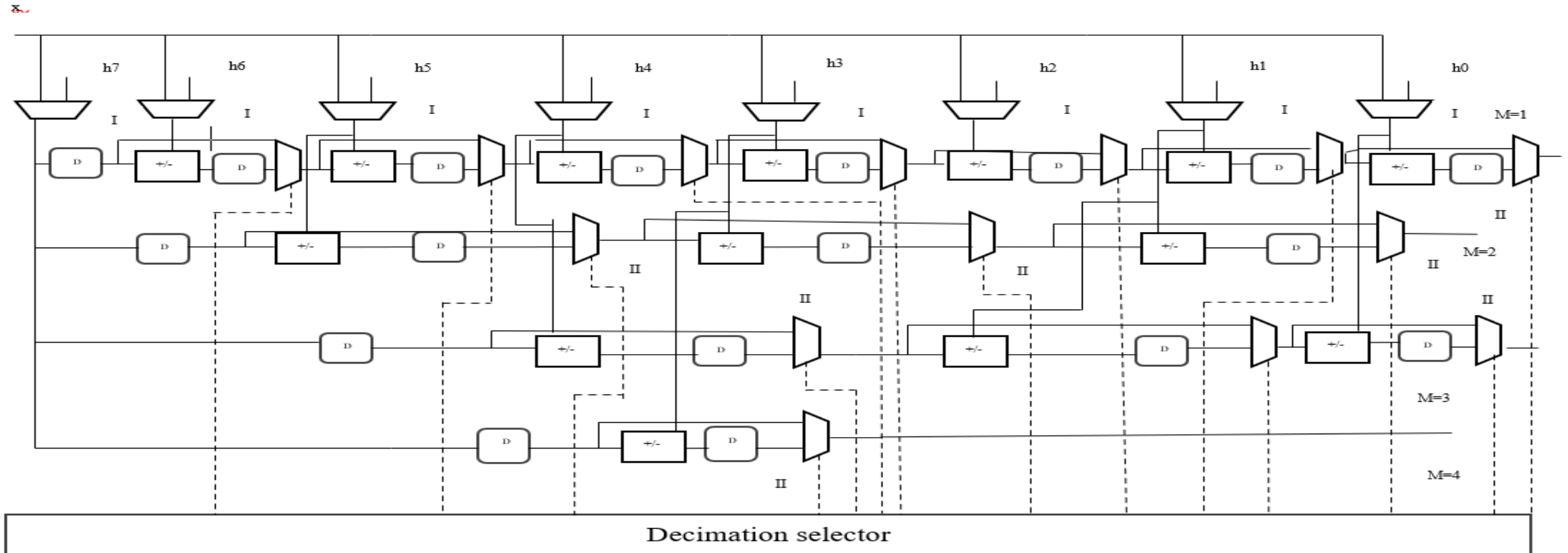


Fig 1.1 shows the ICDM Architecture for 8-Tap filter

Pipelined ICDM Architecture

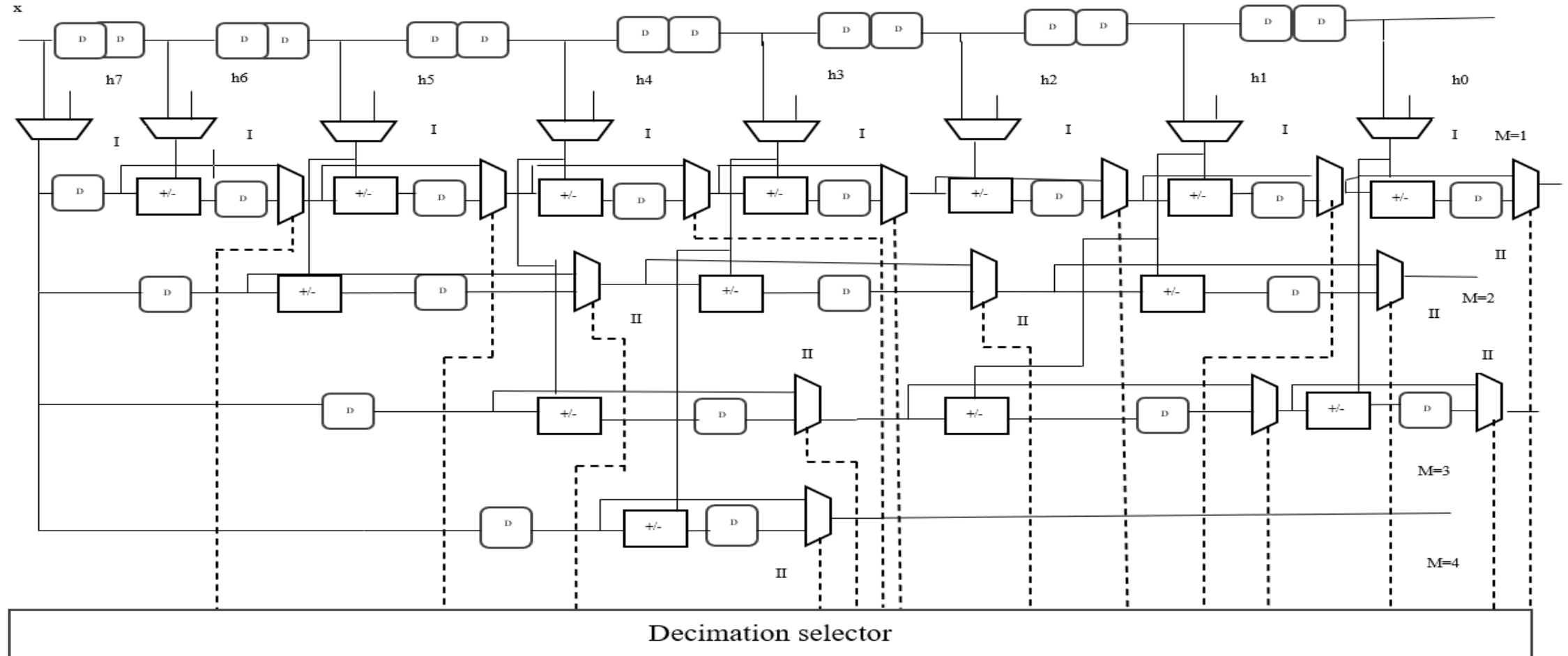


Fig 1.2 shows the Pipelined ICDM Architecture for 8-Tap filter

Coefficient Decimation Method (CDM I)

- Every Mth coefficient is retained and all others are replaced by zeros.

- $h'(n) = h(n)c_M(n)$

where $c_M = 1$; for $n=mM$; $m= 0,1,2$ etc.

$= 0$ otherwise

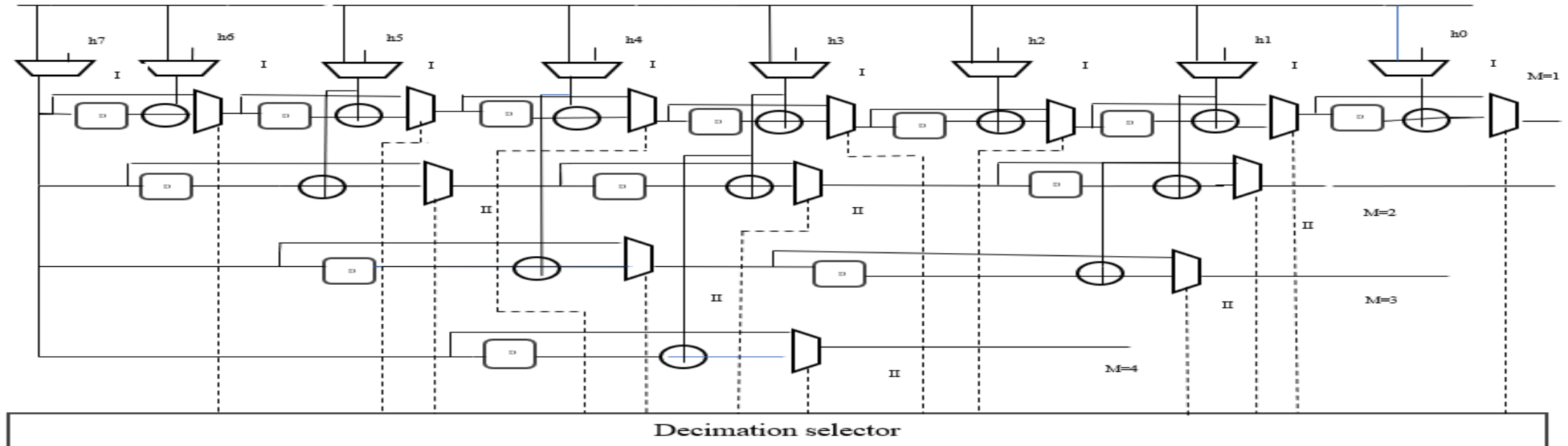


Fig 1.3 shows the Coefficient Decimation Method (CDM I).

Coefficient Decimation Method (CDM II)

- Every M th coefficient is retained and all others value are discarded.
- To obtained low pass frequency response.

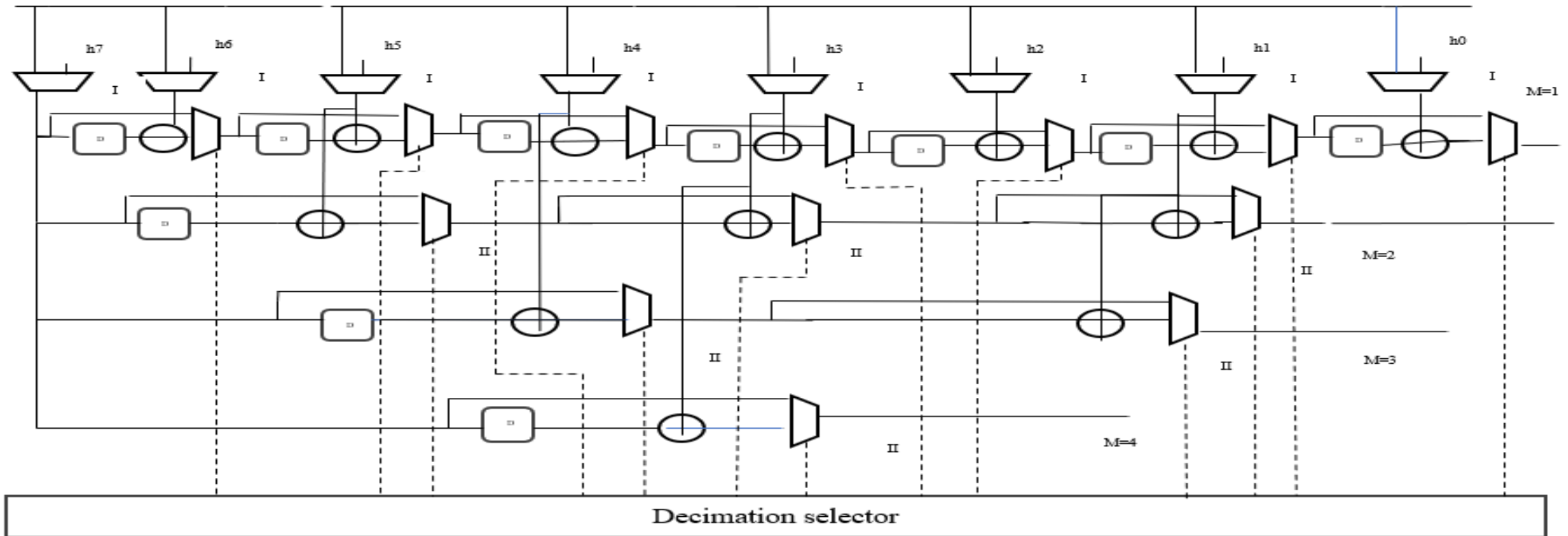


Fig 1.4 shows the Coefficient Decimation Method (CDM II).

Modified Coefficient Decimation Method (MCDM I)

- Every Mth coefficient is retained and sign of every alternate retained coefficient is reversed . All others are replaced by zeros.
- $h'(n) = h(n)d_M(n)$

where $d_M = 1$; for $n=mM$; $m= 0,2,4,6 \dots$

$= -1$; for $n=pM$; $p= 1,3,5,7 \dots$

$= 0$ otherwise

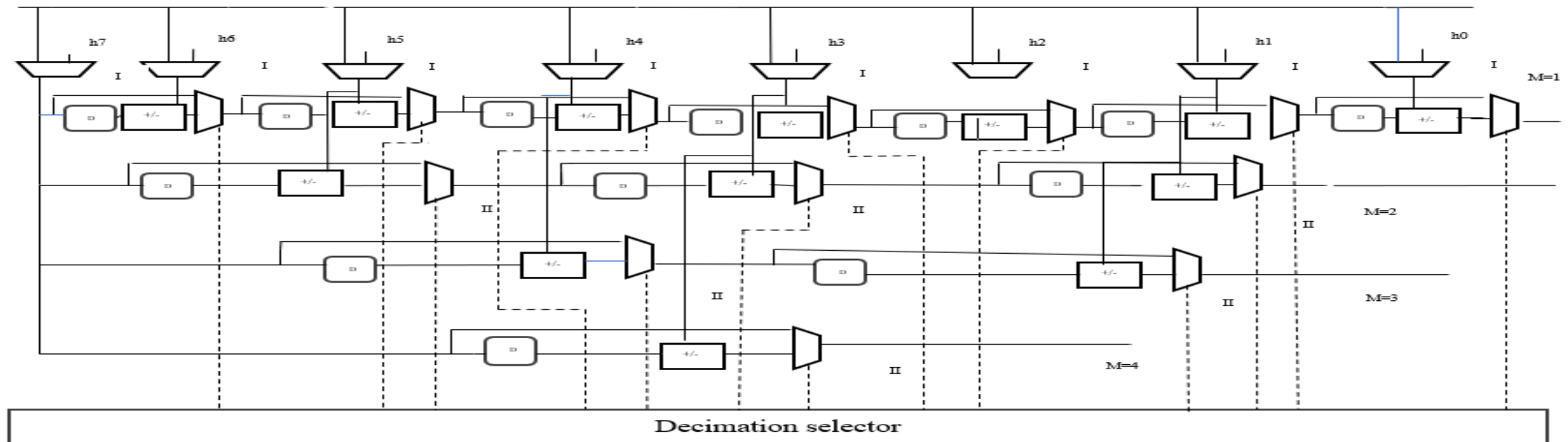


Fig1.5 shows the Modified Coefficient Decimation Method (MCDM I)

Modified Coefficient Decimation Method (MCDM II)

- Every Mth coefficient is retained and sign of every alternate retained coefficient is reversed. All others values are discarded.

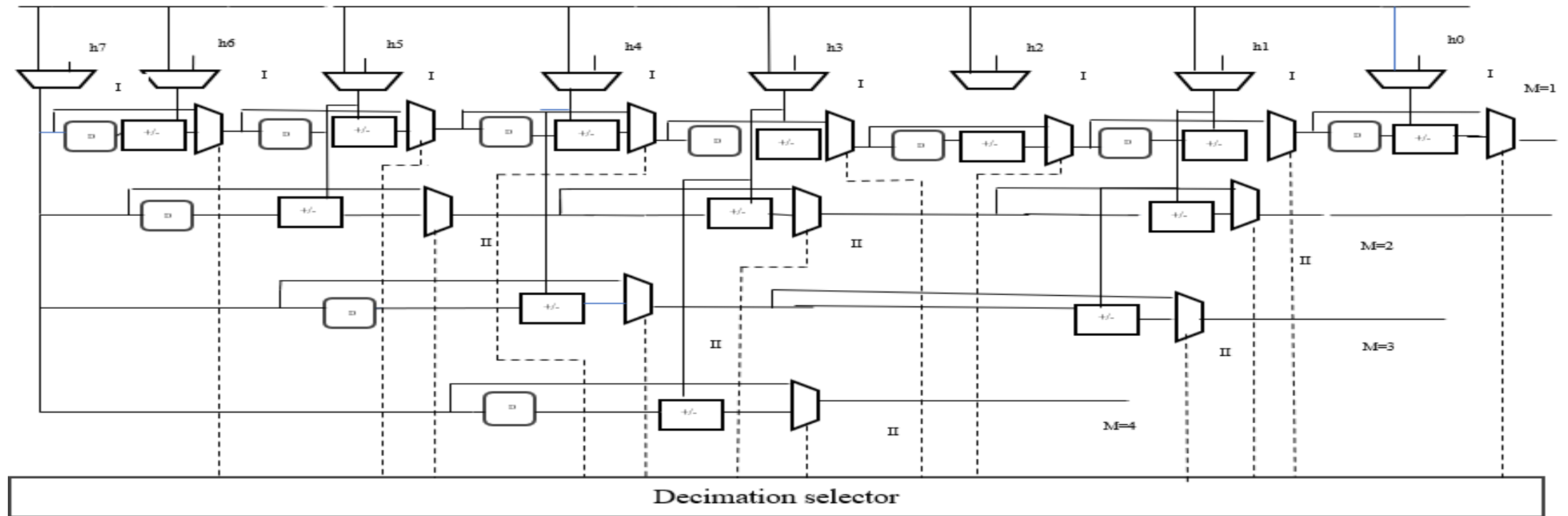


Fig1.6 shows the Modified Coefficient Decimation Method (MCDM II)

Improved Coefficient Decimation Method(ICDM I)

- Combination of CDM I and MCDM I is called Improved Coefficient Decimation Method I (ICDM I).

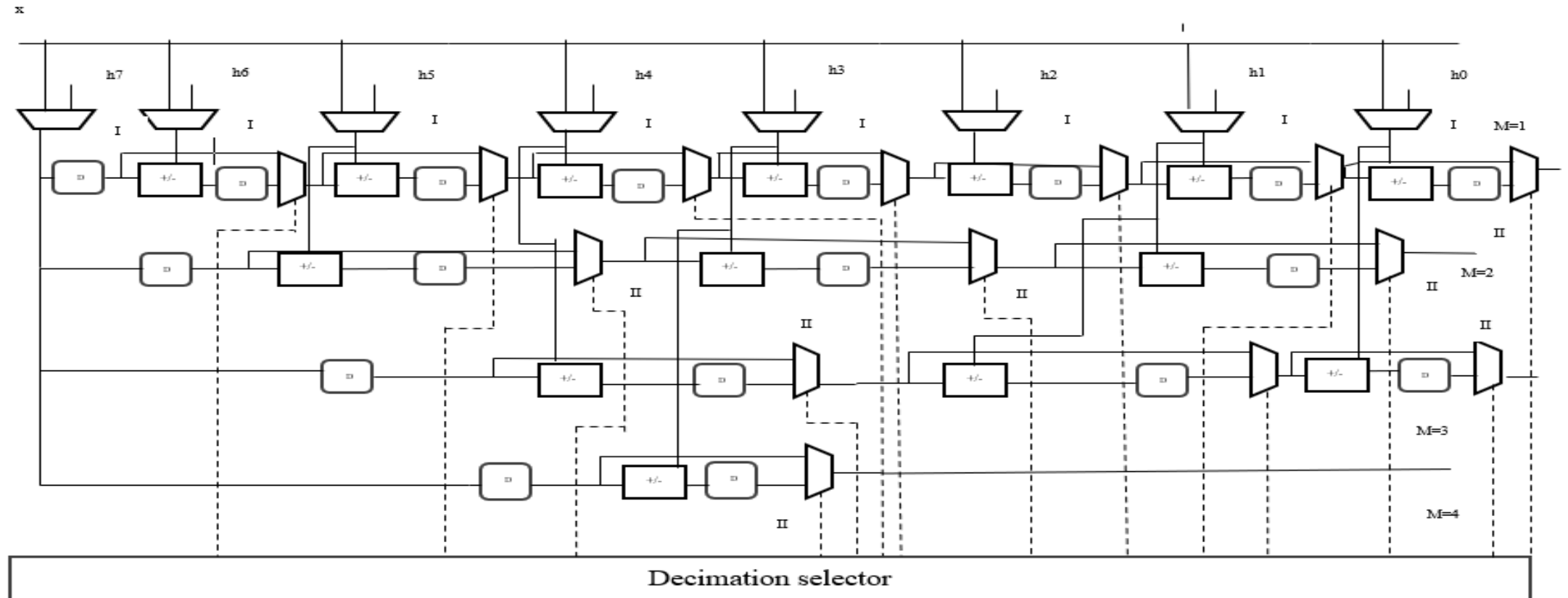


Fig 1.7 shows the Improved Coefficient Decimation Method (ICDM I)

Improved Coefficient Decimation Method(ICDM II)

- Combination of CDM II and MCDM II is called Improved Coefficient Decimation Method II (ICDM II).

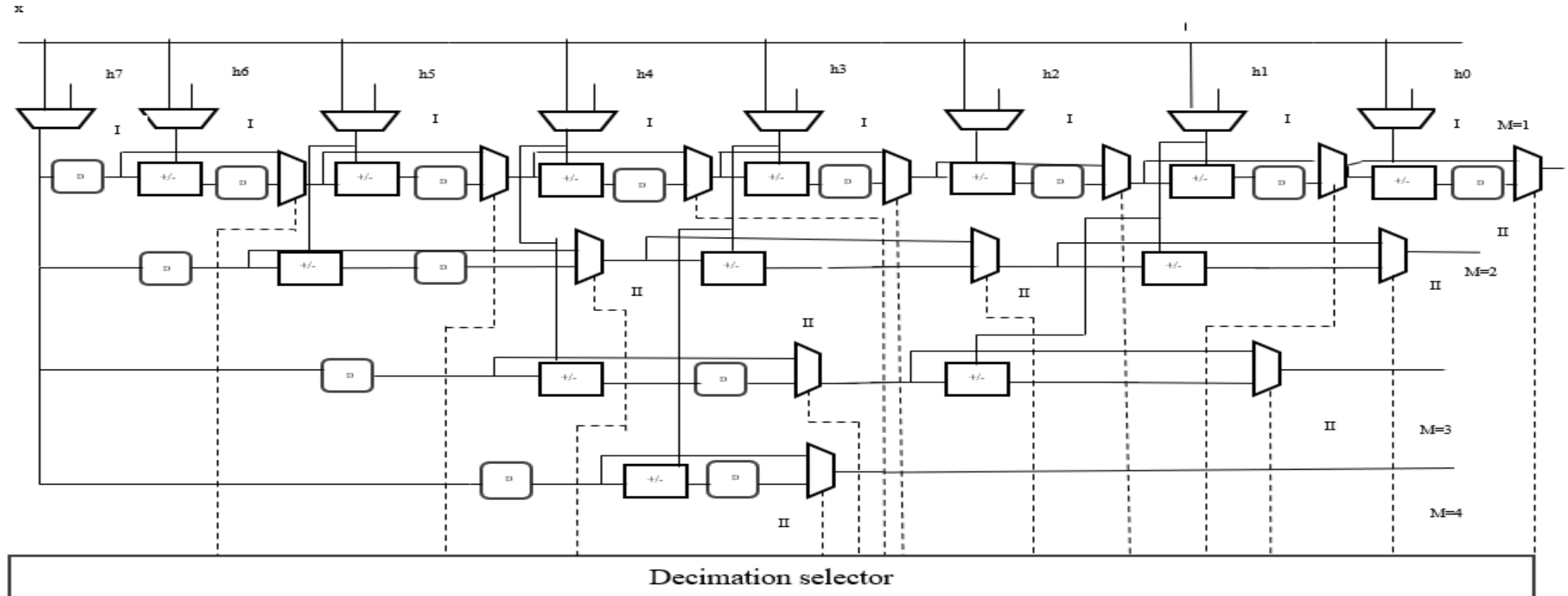
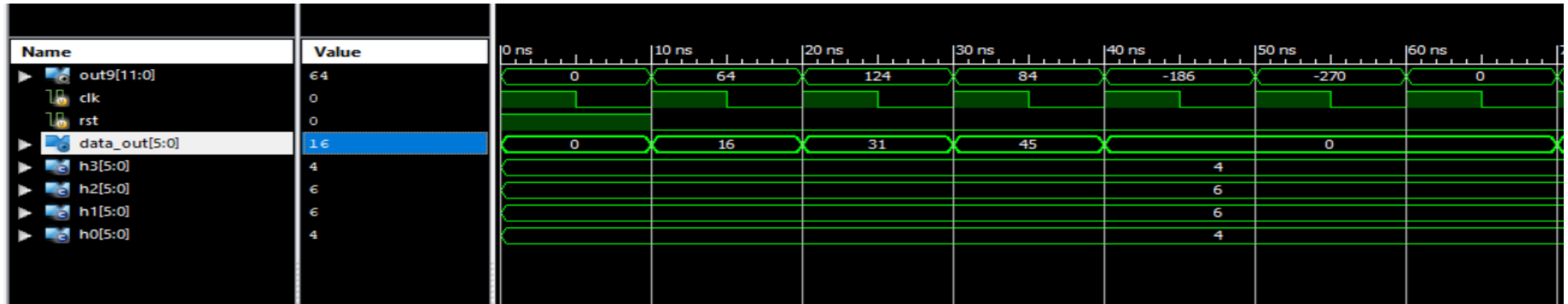


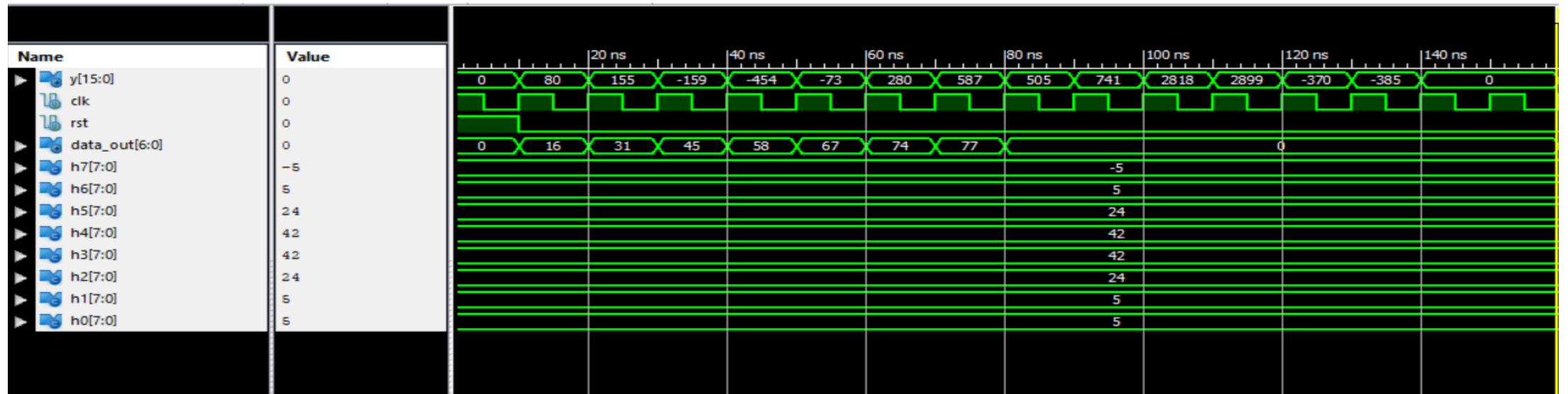
Fig 1.8 shows the Improved Coefficient Decimation Method (ICDM II)

Simulation Waveform

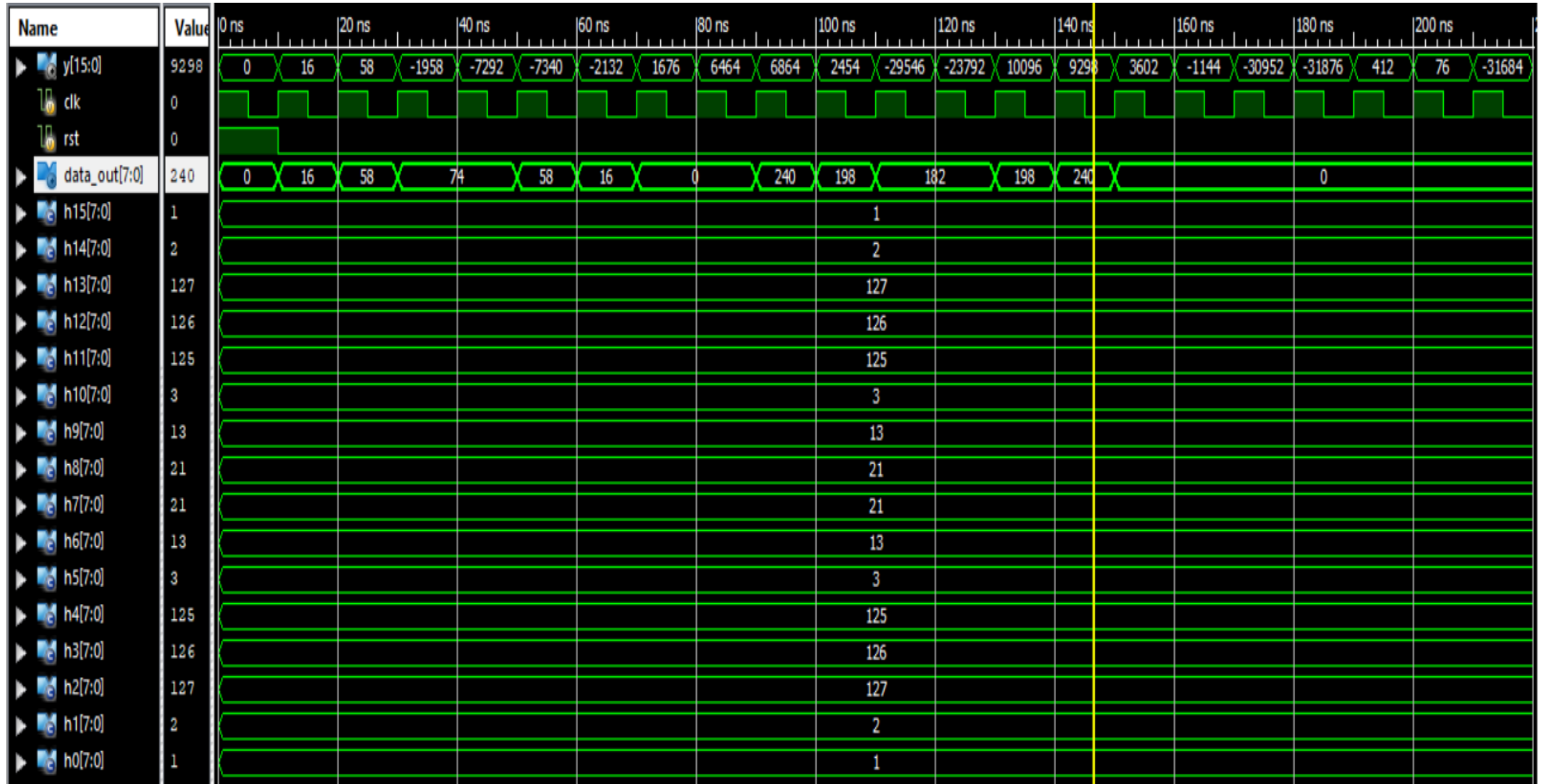
- 4 Tap filter for ICDM I



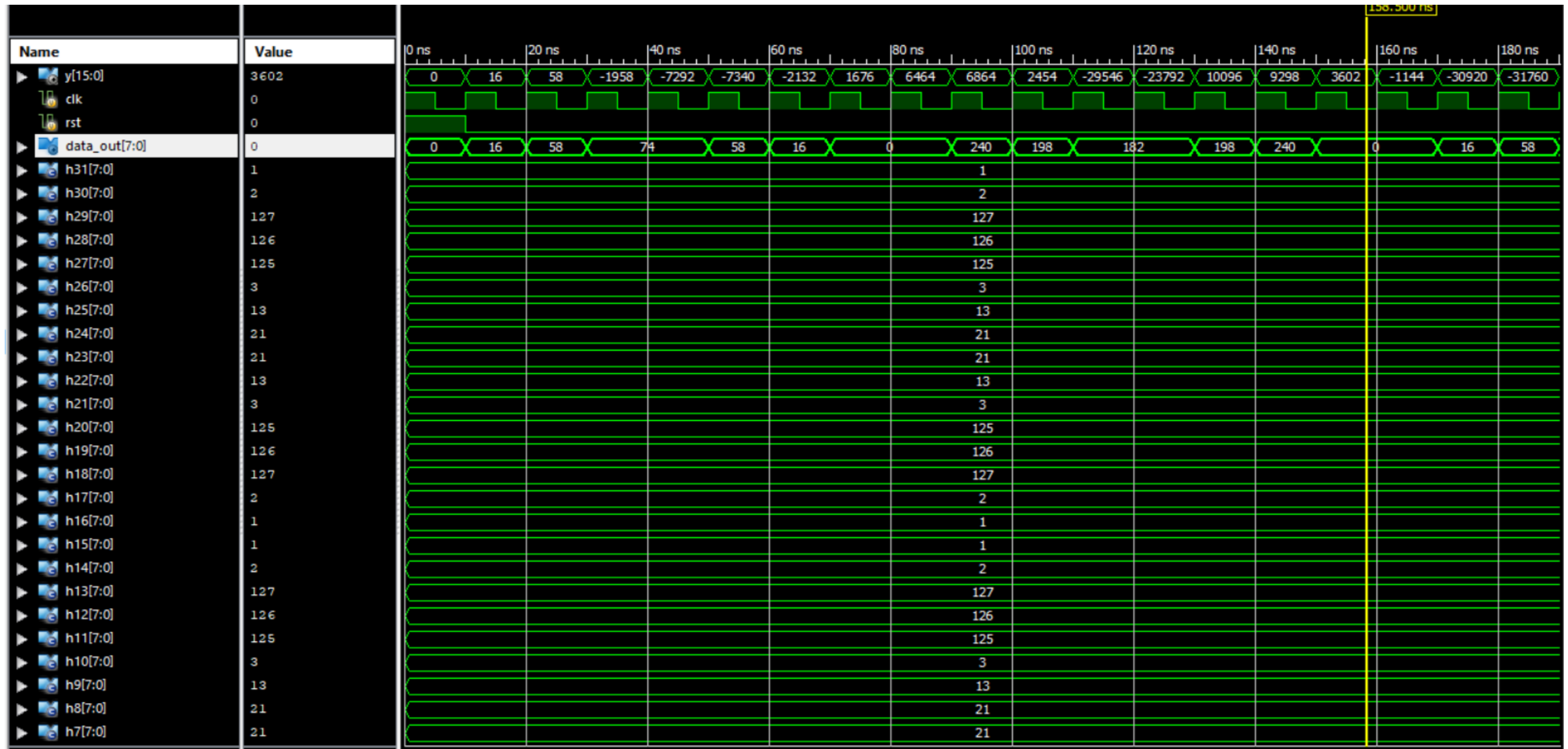
- 8 Tap filter for ICDM I



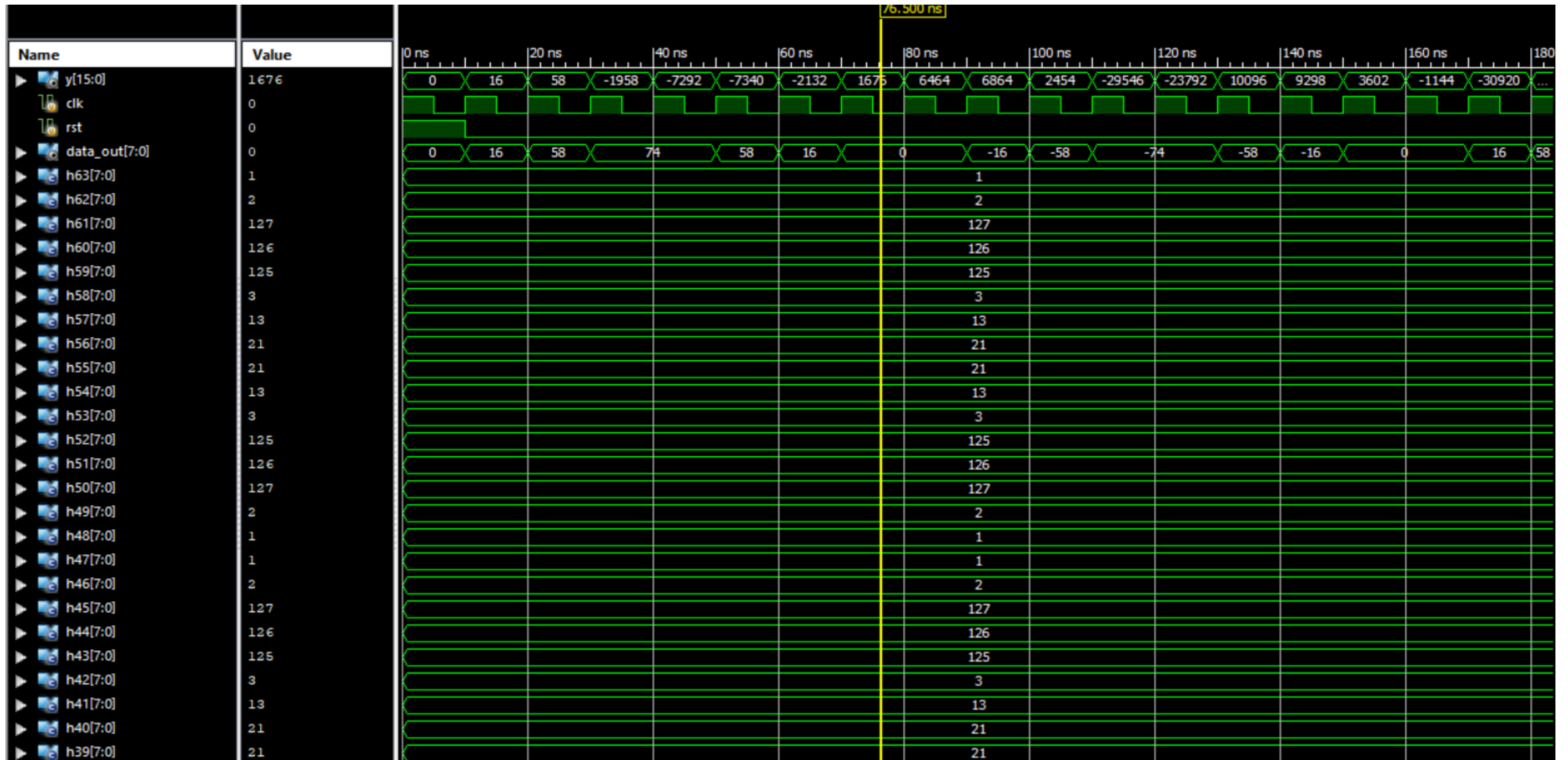
- 16 Tap filter for ICDM I



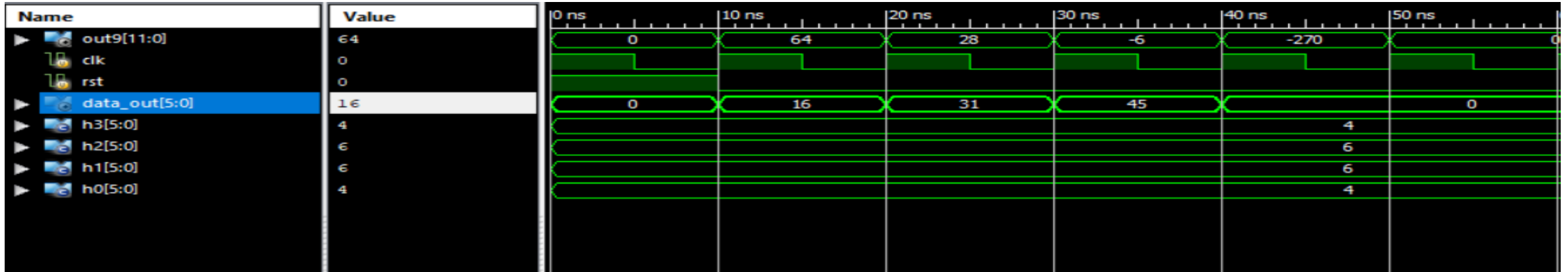
- 32 Tap filter for ICDM I



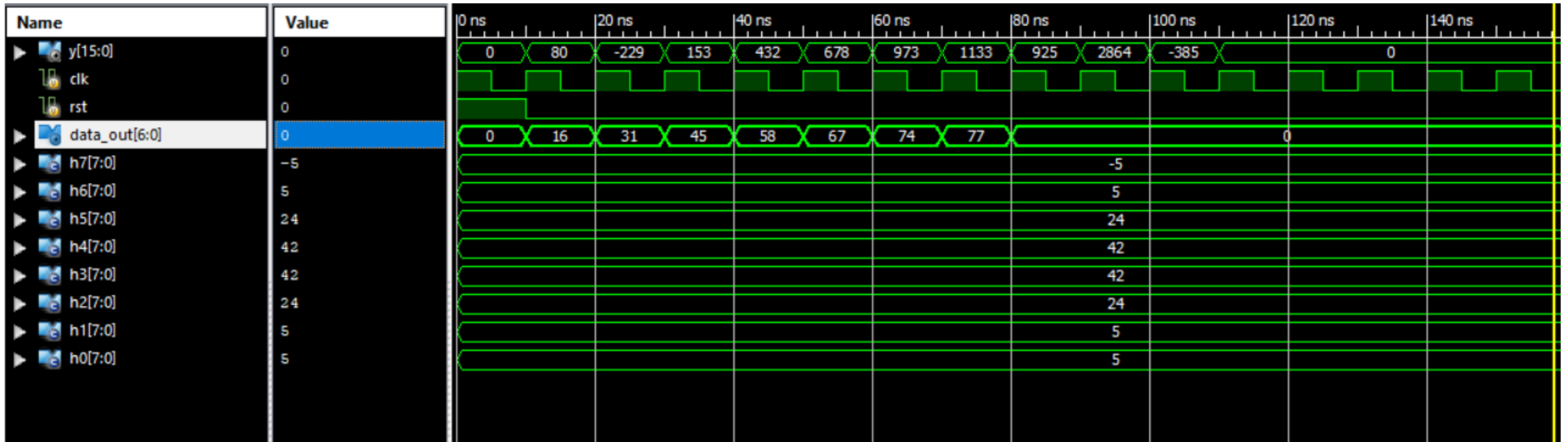
- 64 Tap filter for ICDM I



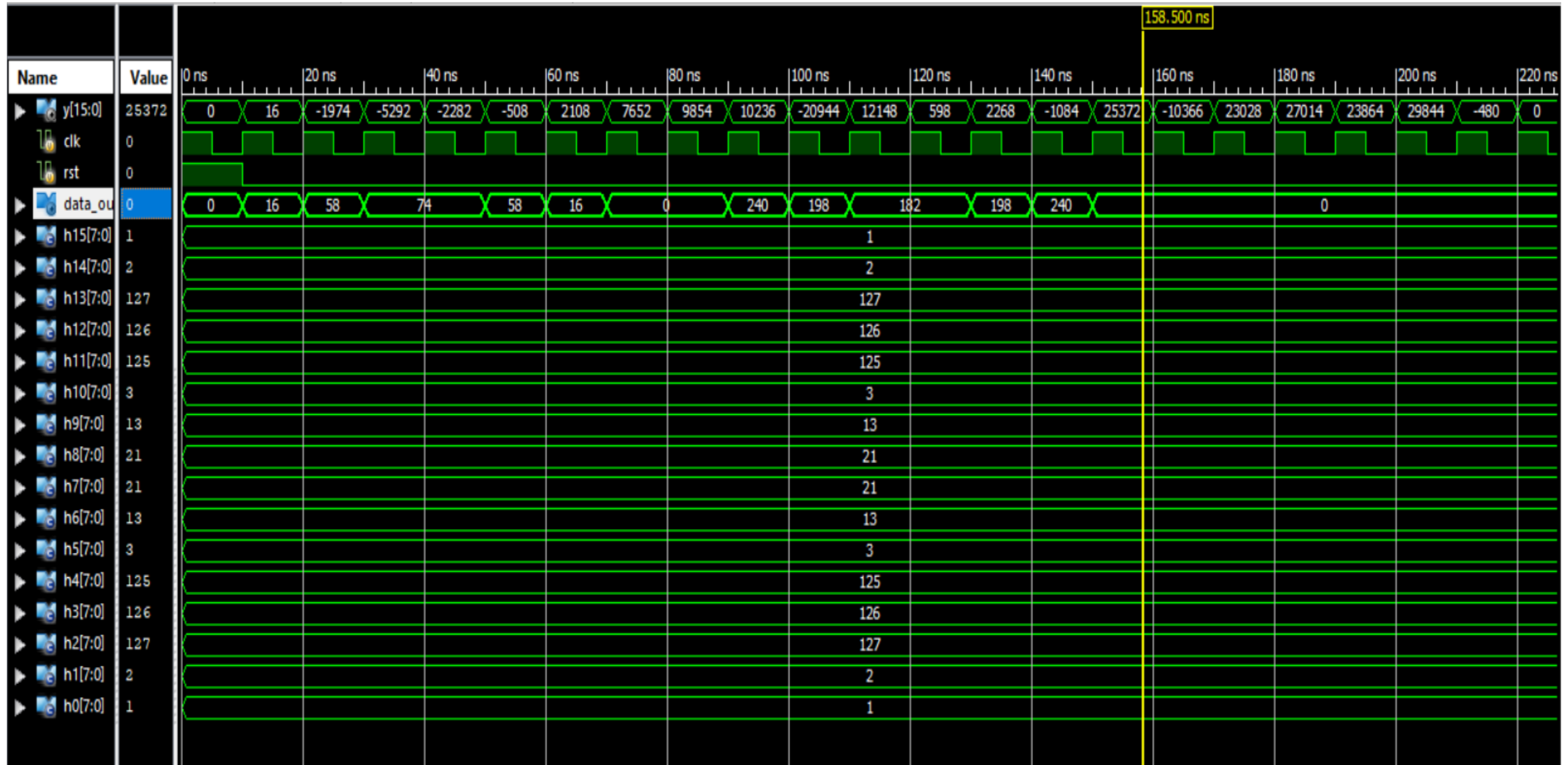
- 4 Tap filter for ICDM II



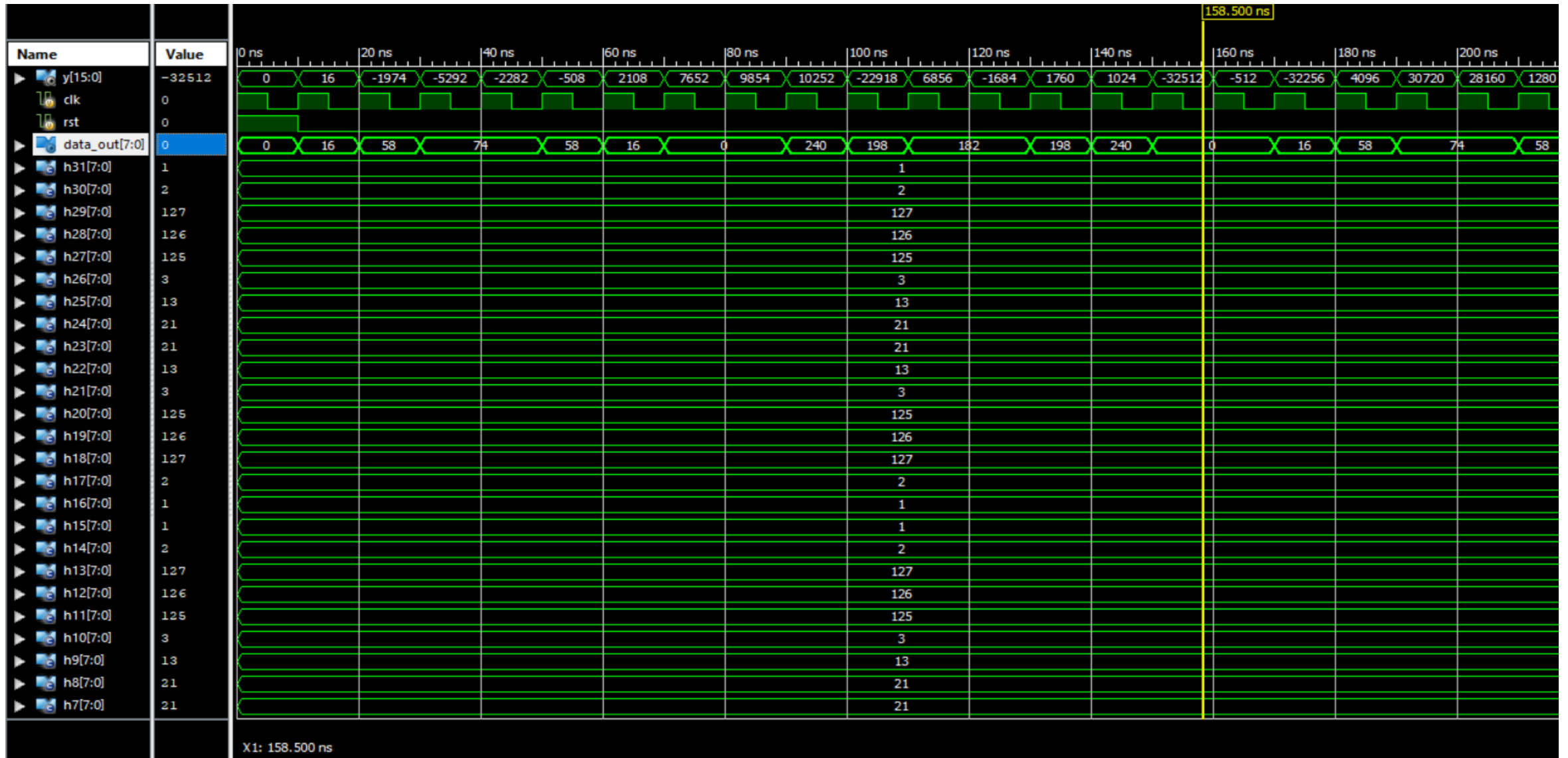
- 8 Tap filter for ICDM II



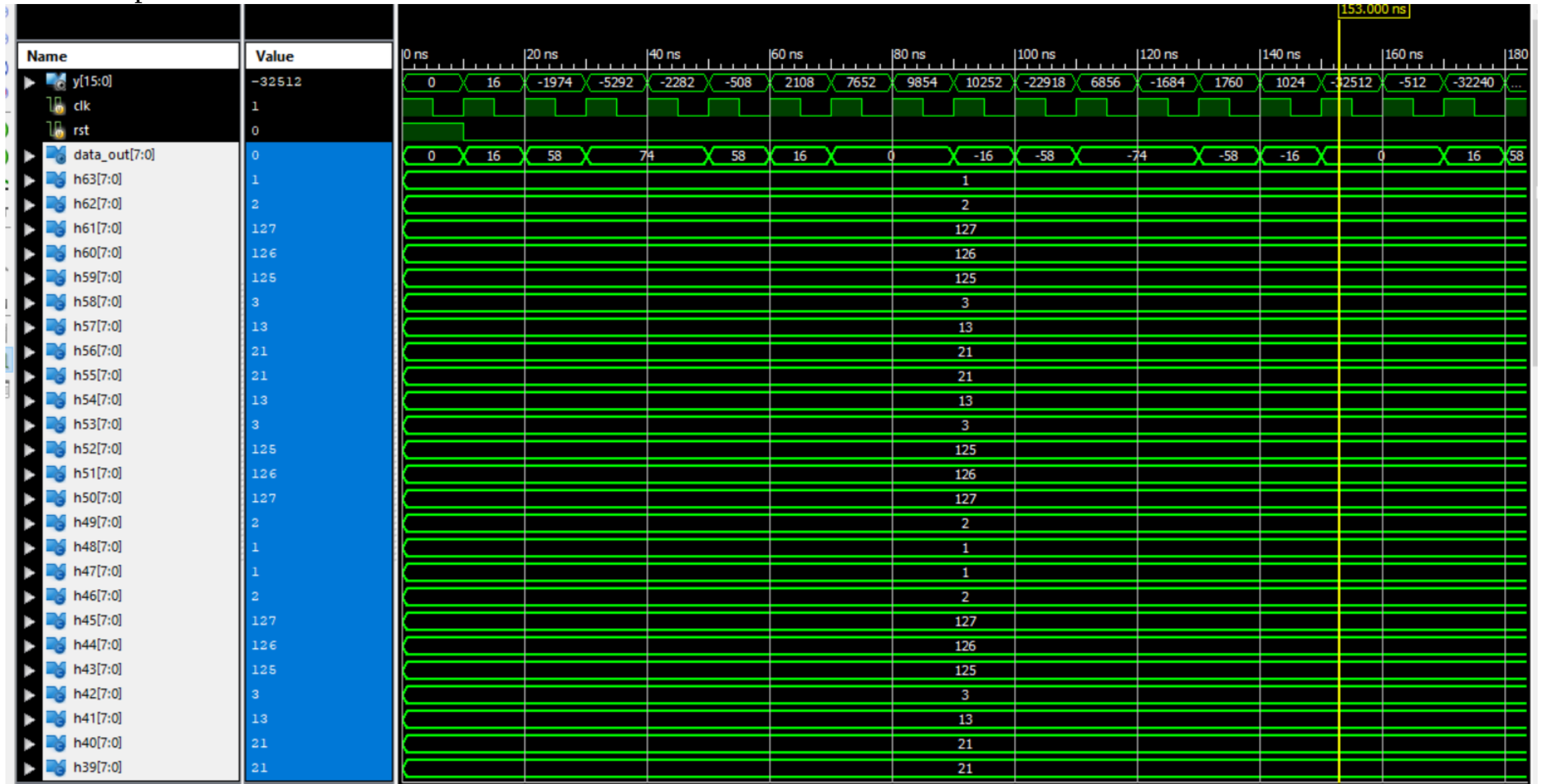
- 16 Tap filter for ICDM II



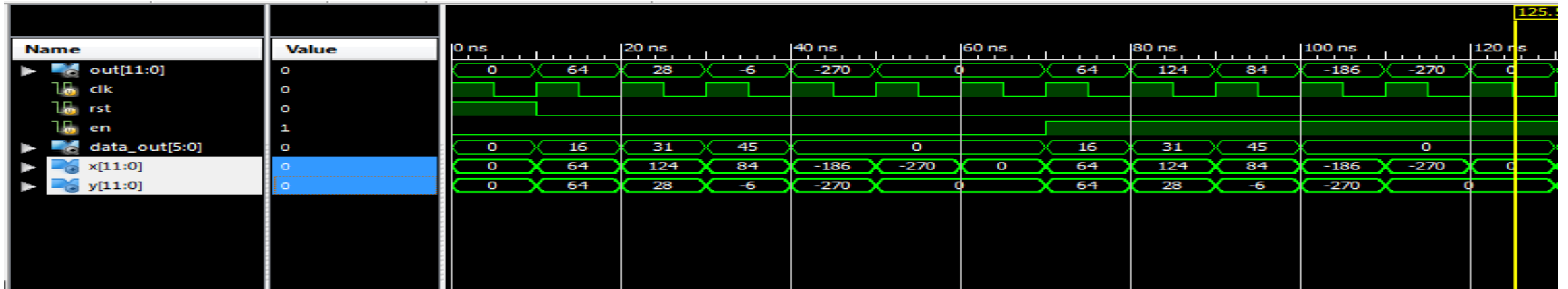
- 32 Tap Filter for ICDM II



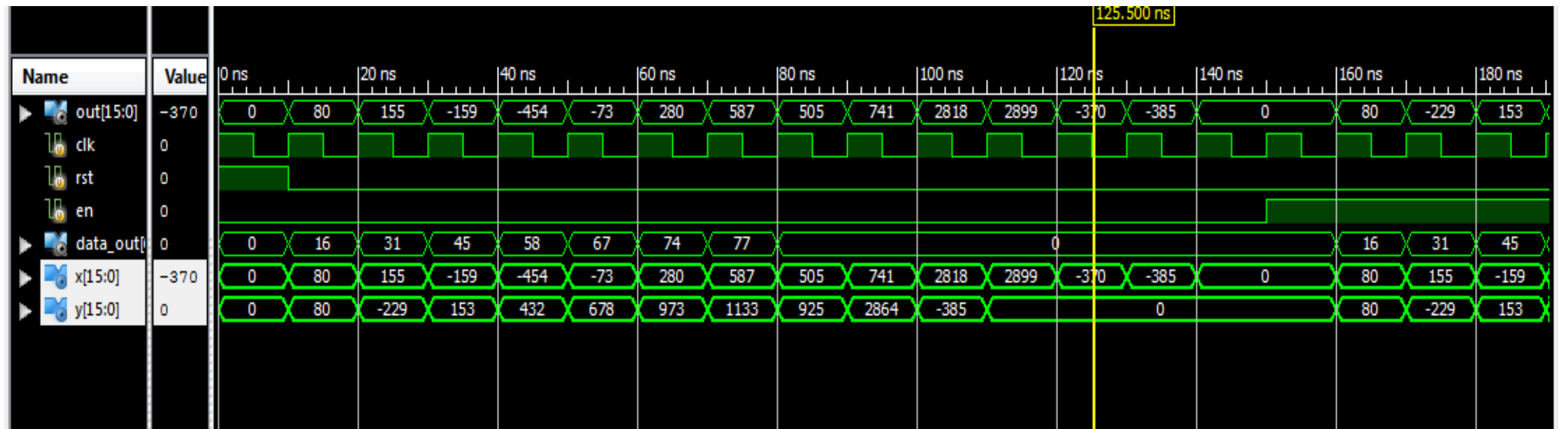
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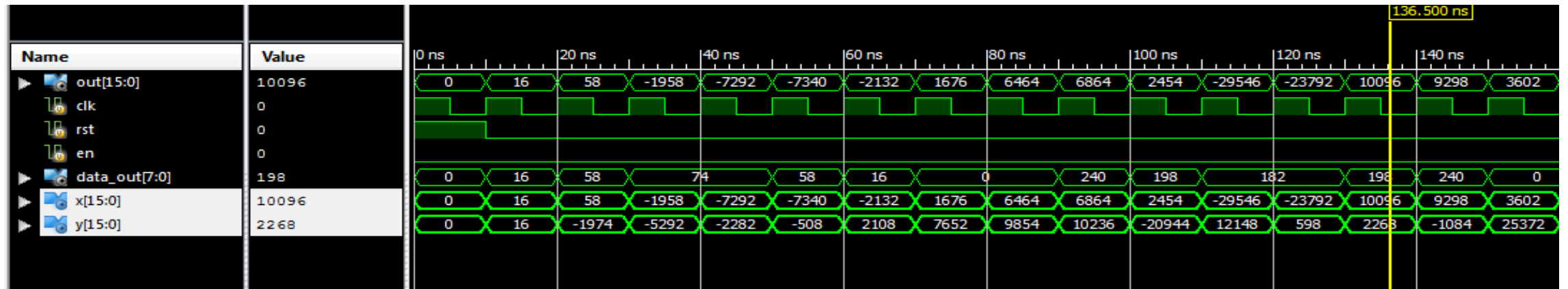
- 4 Tap Filter for ICDM



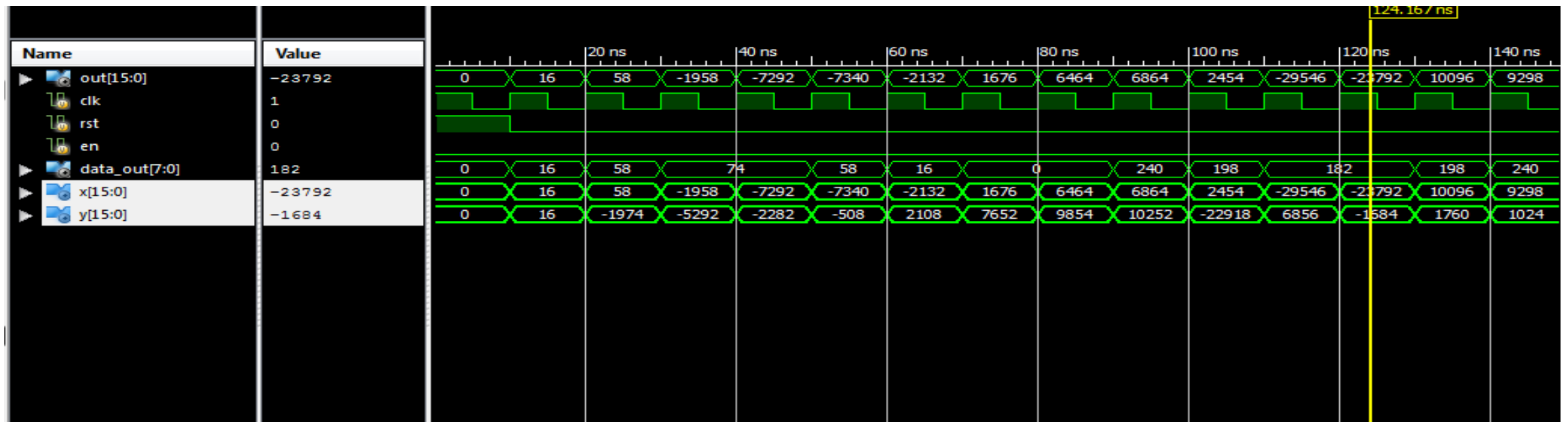
- 8 Tap Filter for ICDM



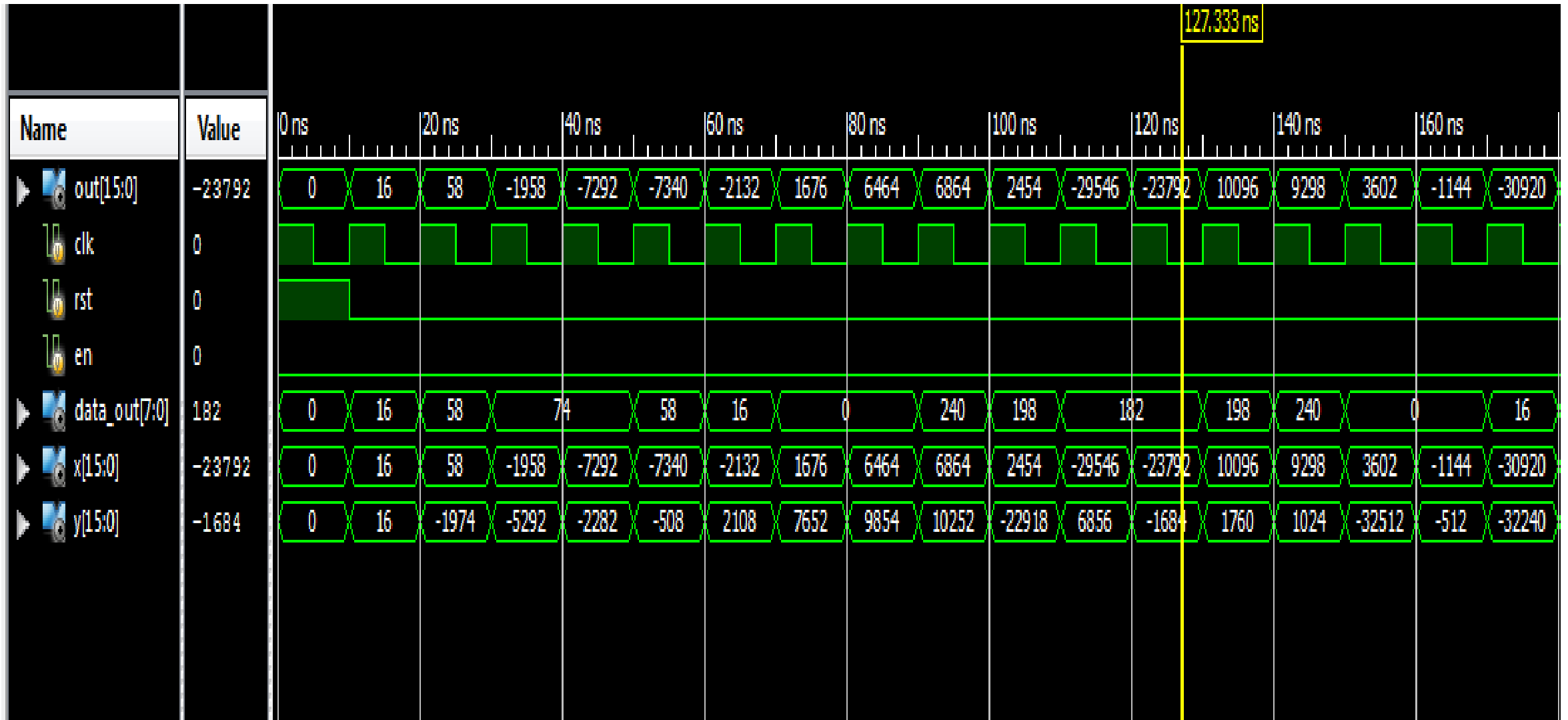
- 16 Tap Filter for ICDM



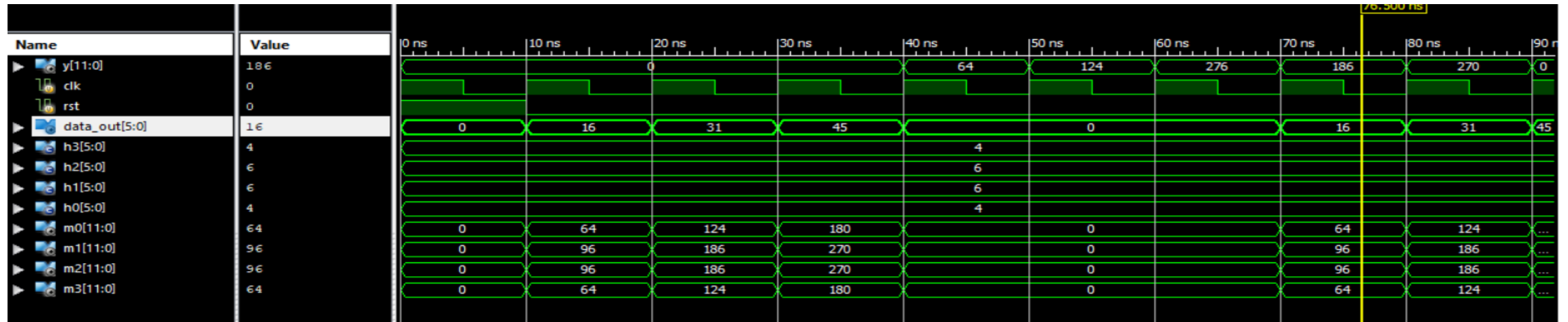
- 32 Tap Filter for ICDM



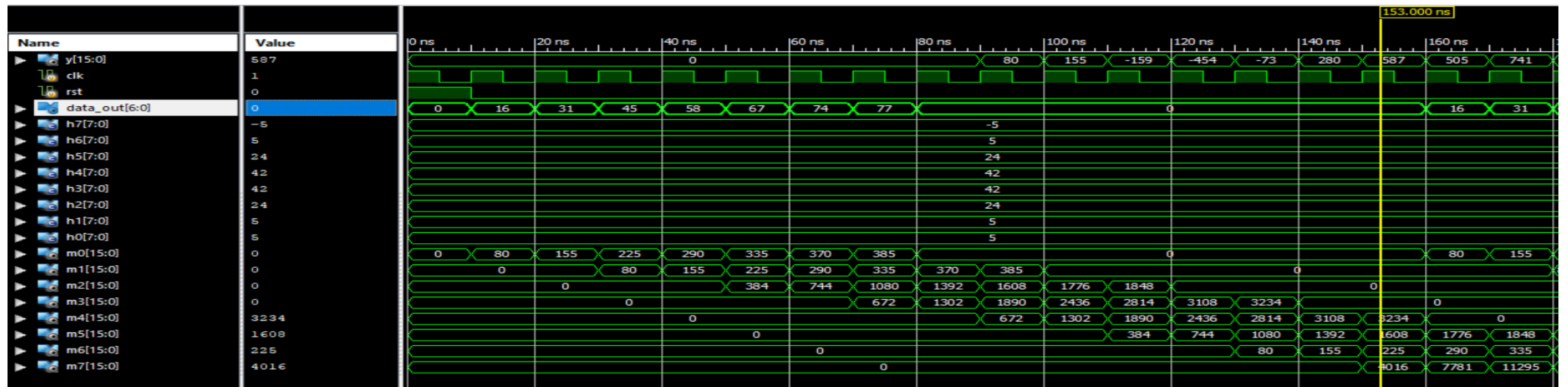
- 64 Tap Filter for ICDM II



- 4 Tap Filter for Pipelined ICDM I



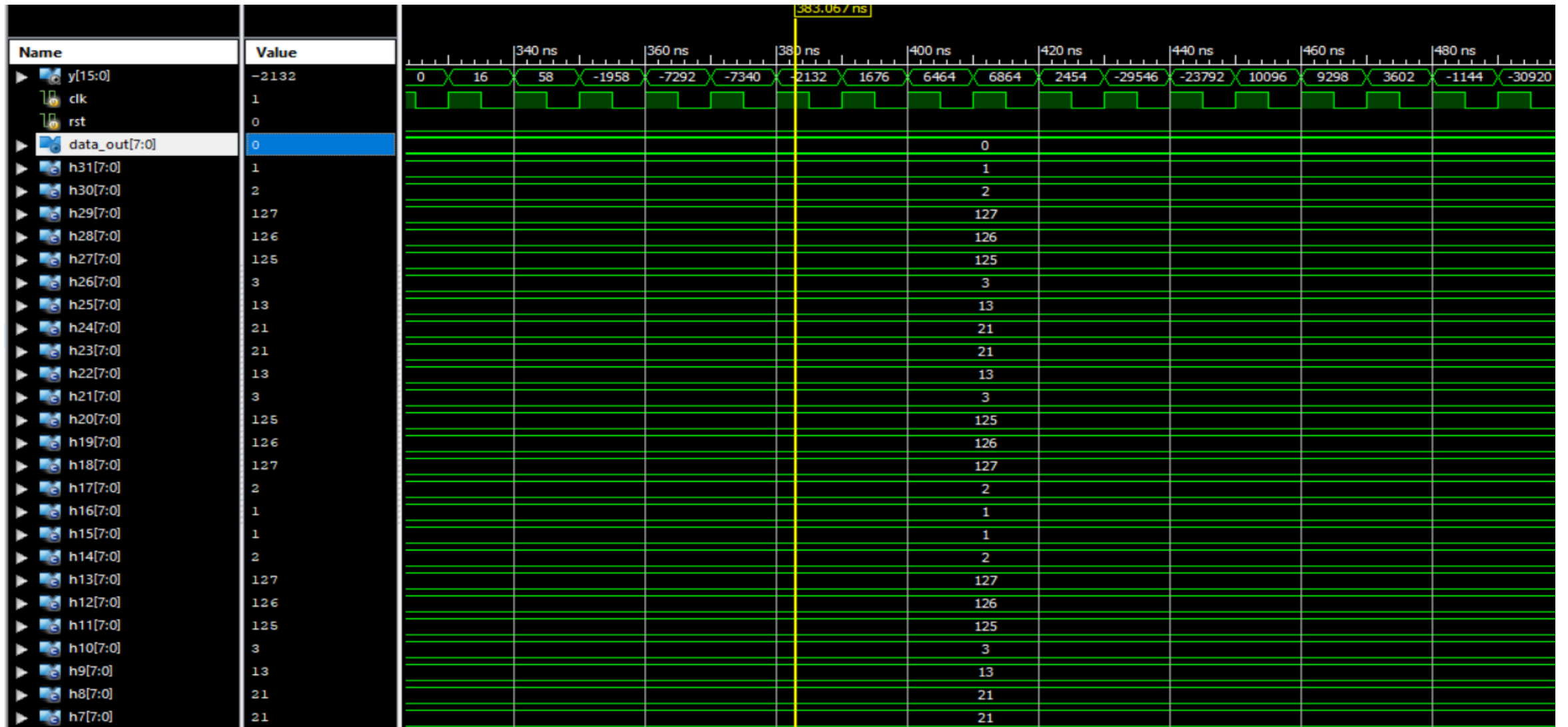
- 8 Tap Filter for Pipelined ICDM I



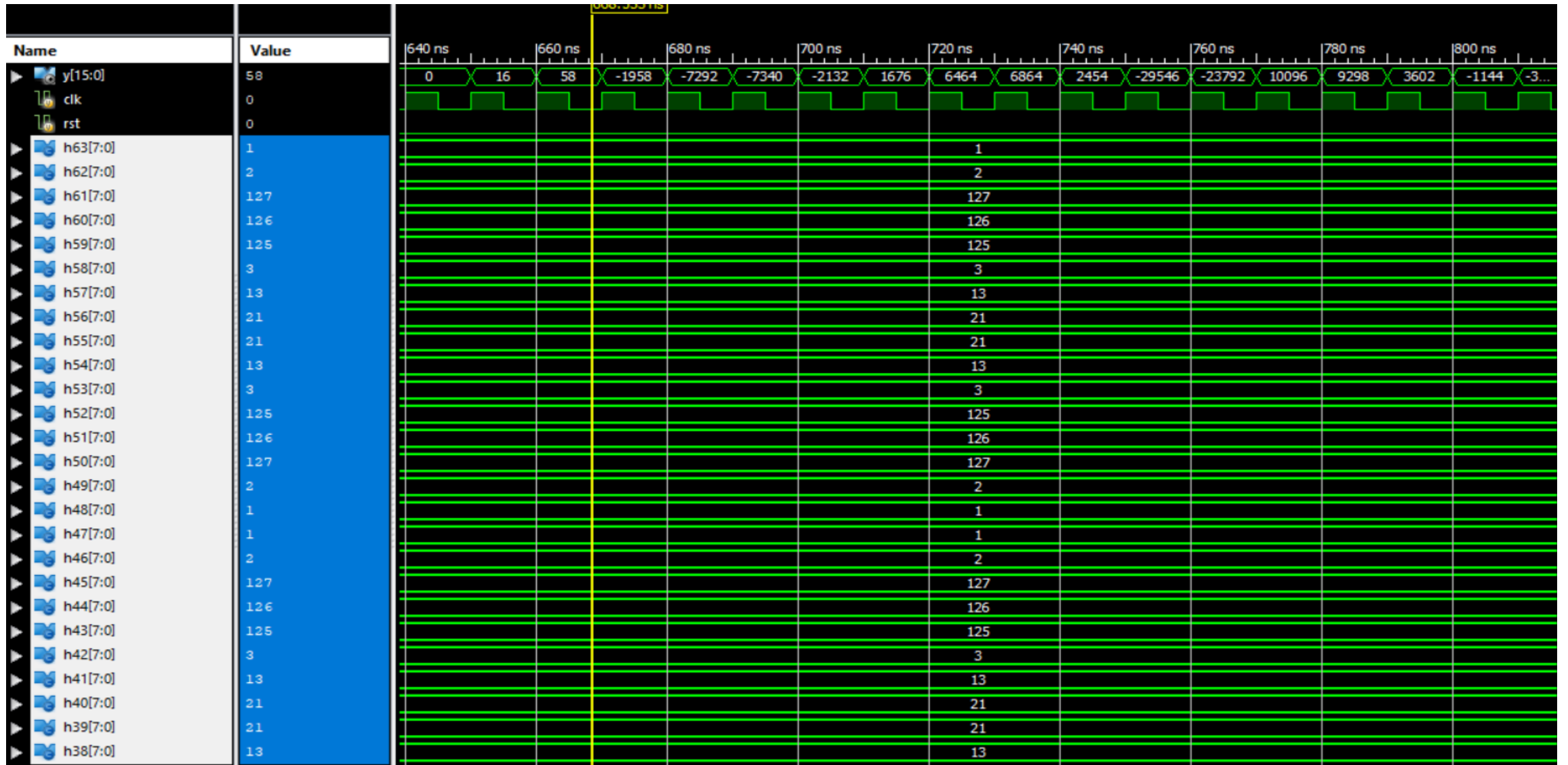
- 16 Tap Filter for Pipelined ICDM I



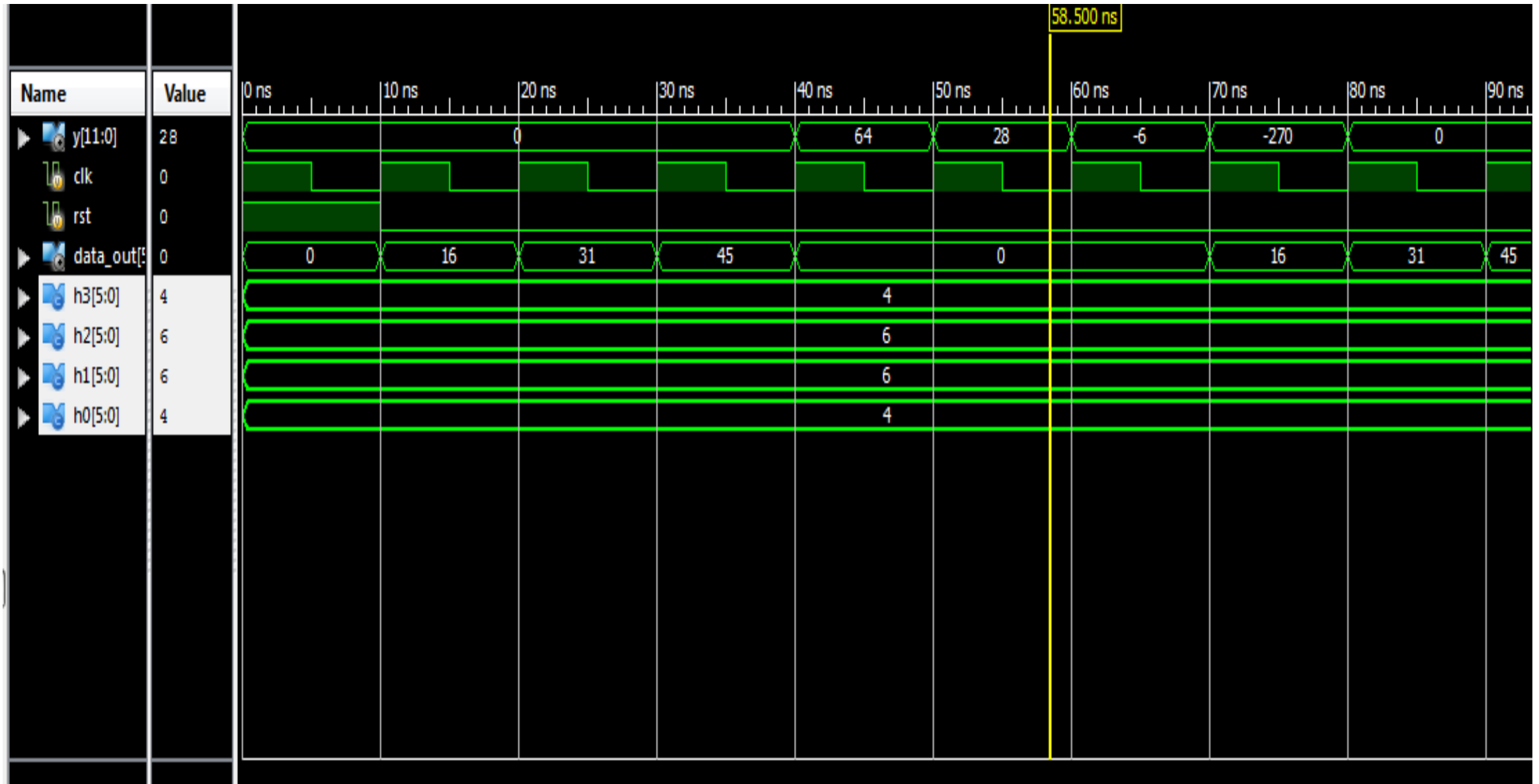
- 32 Tap Filter for Pipelined ICDM I



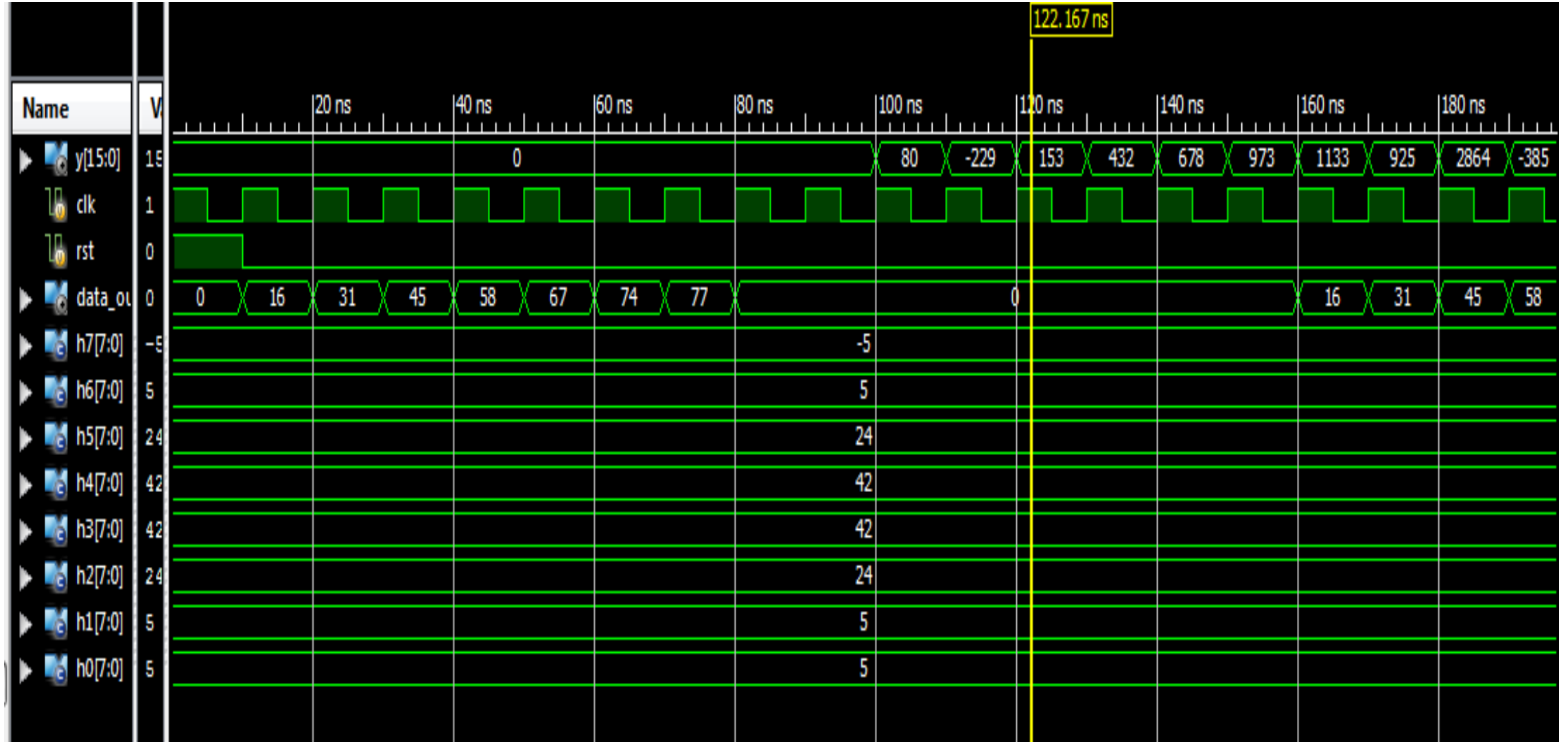
- 64 Tap Filter for Pipelined ICDM I



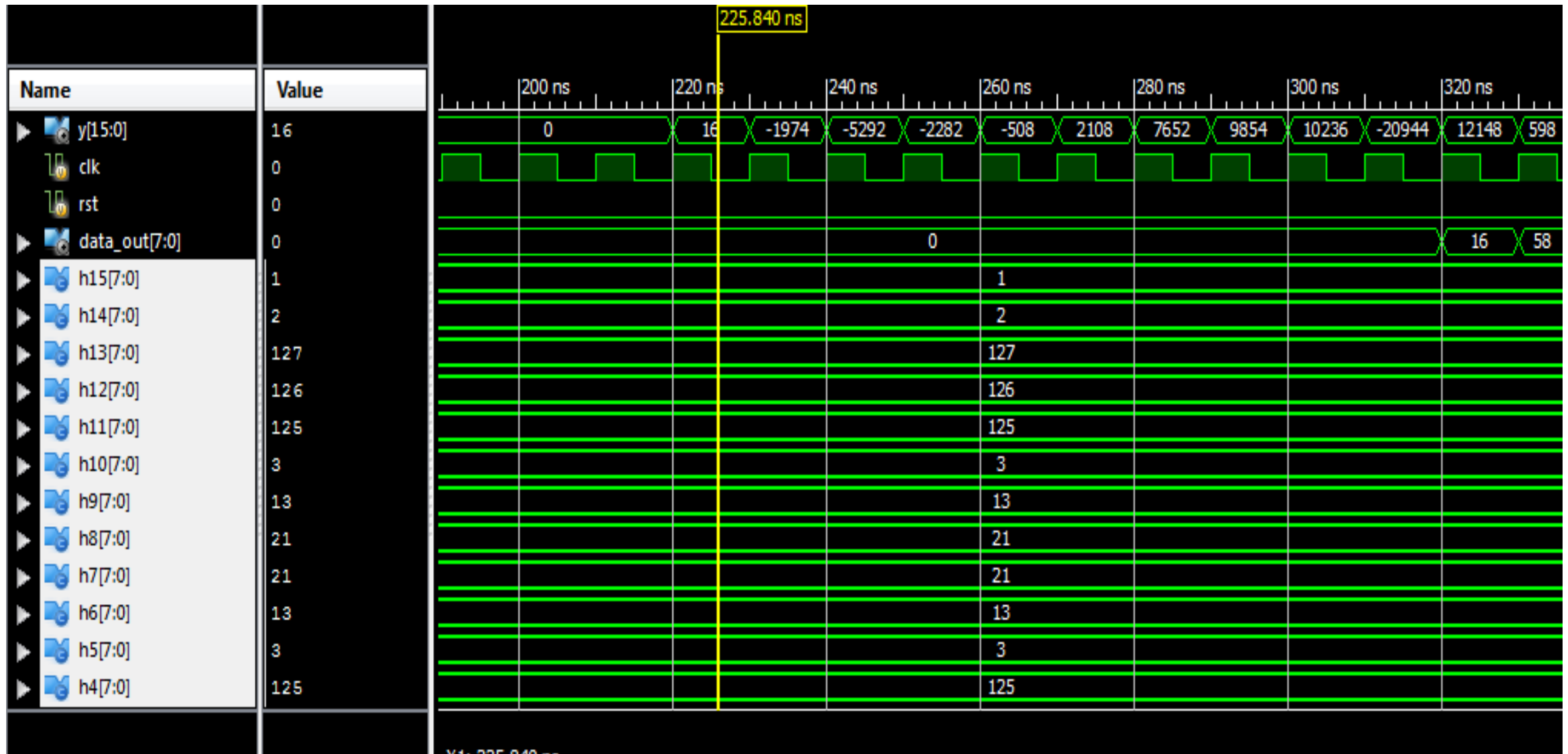
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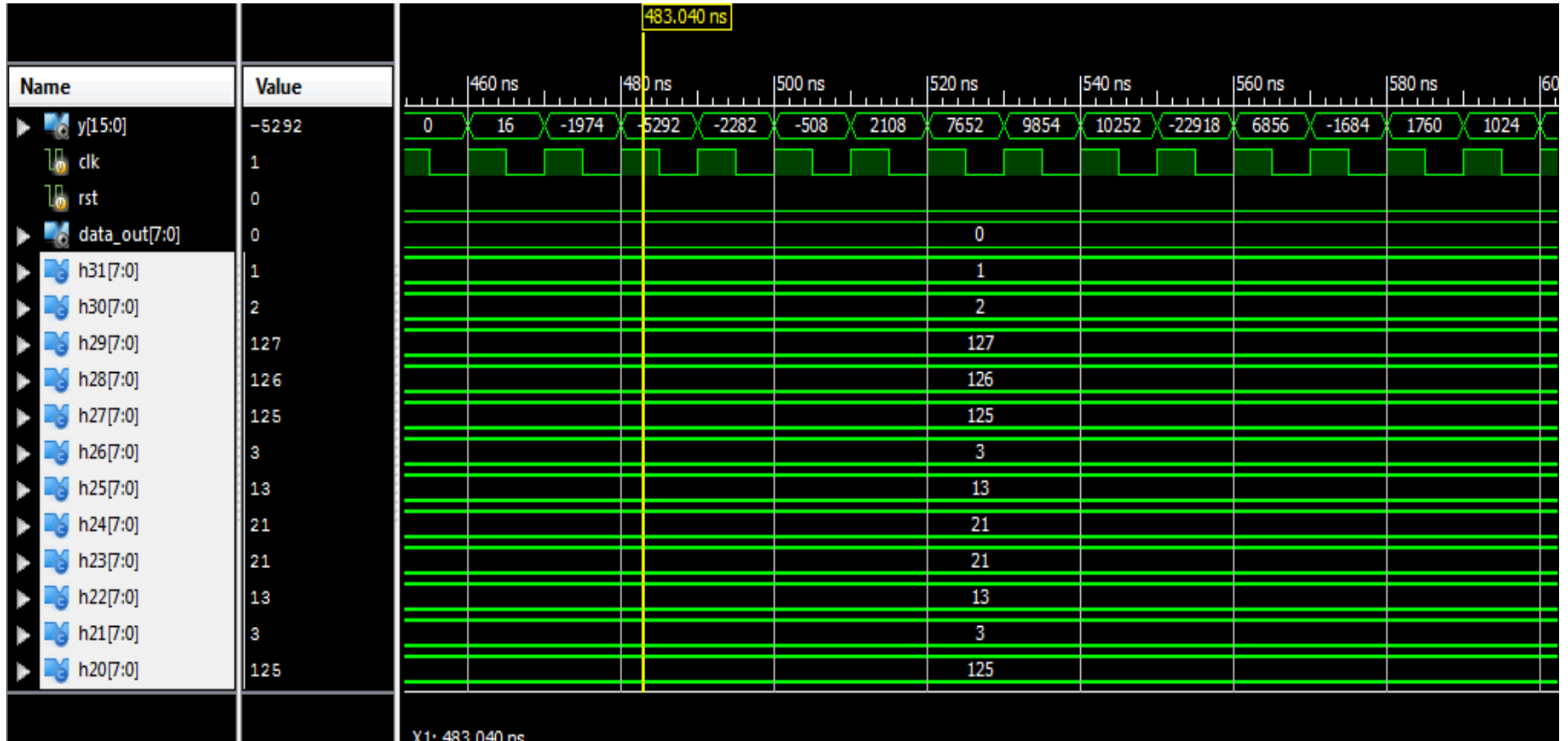
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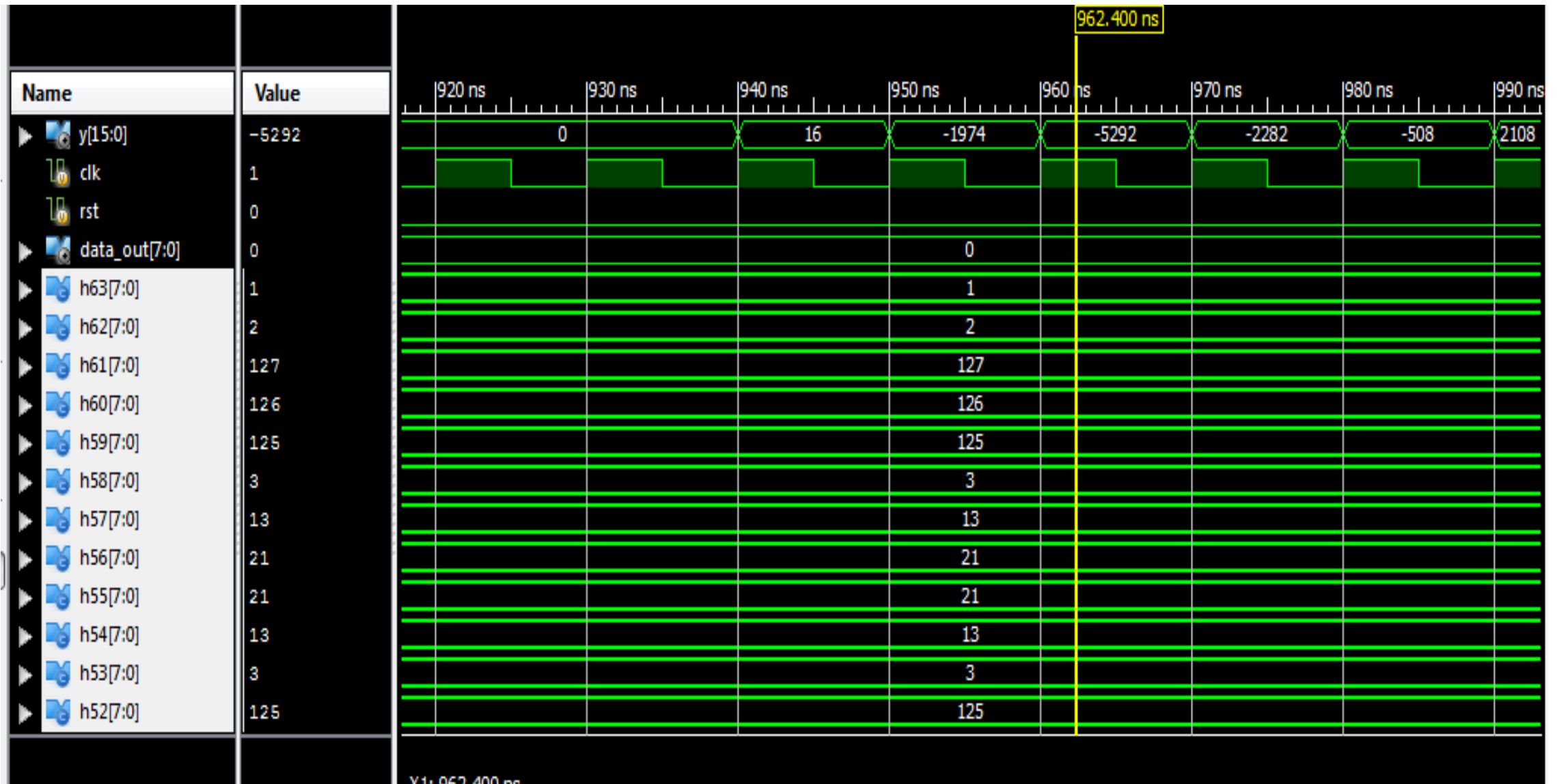
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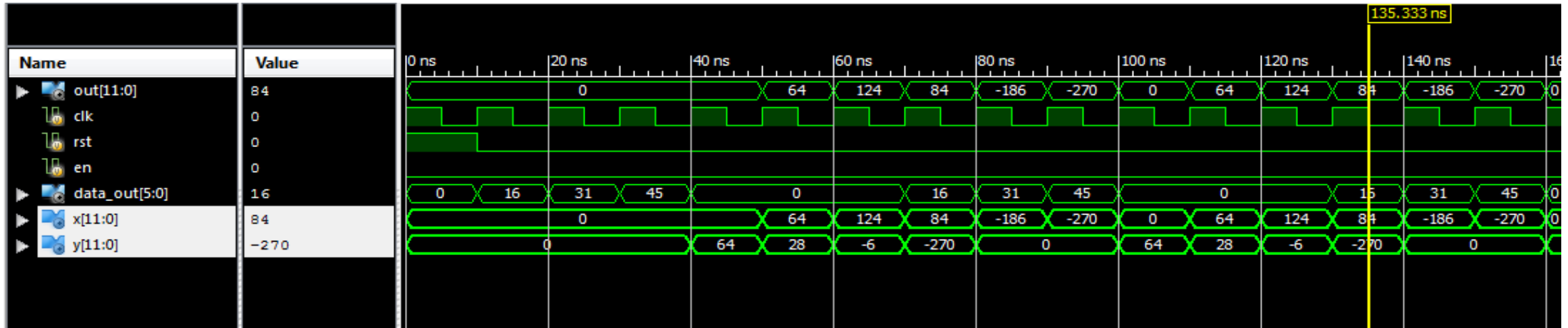
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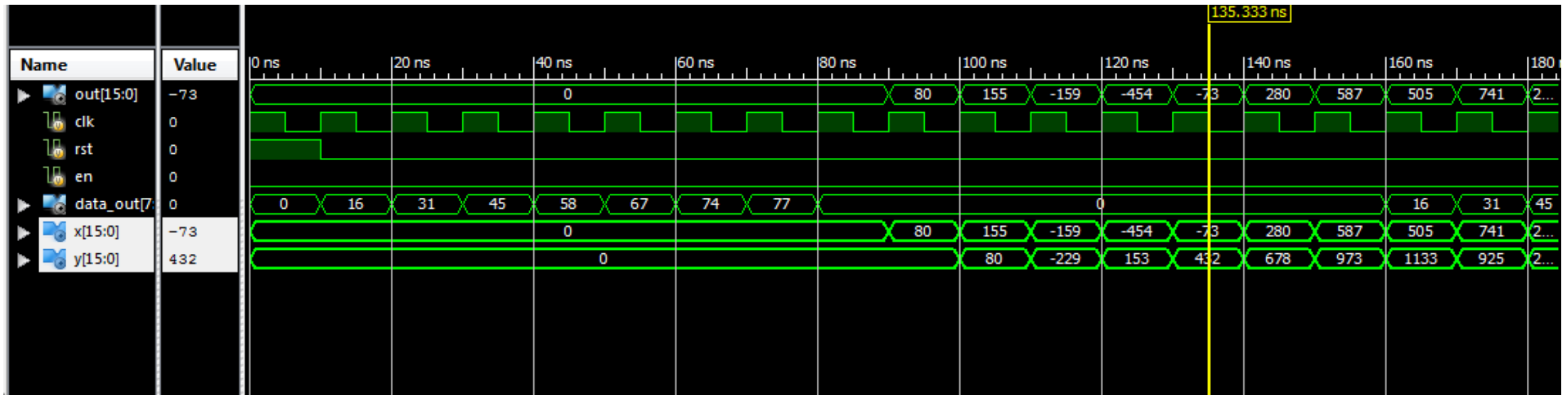
- 64 Tap Filter for Pipelined ICDM II



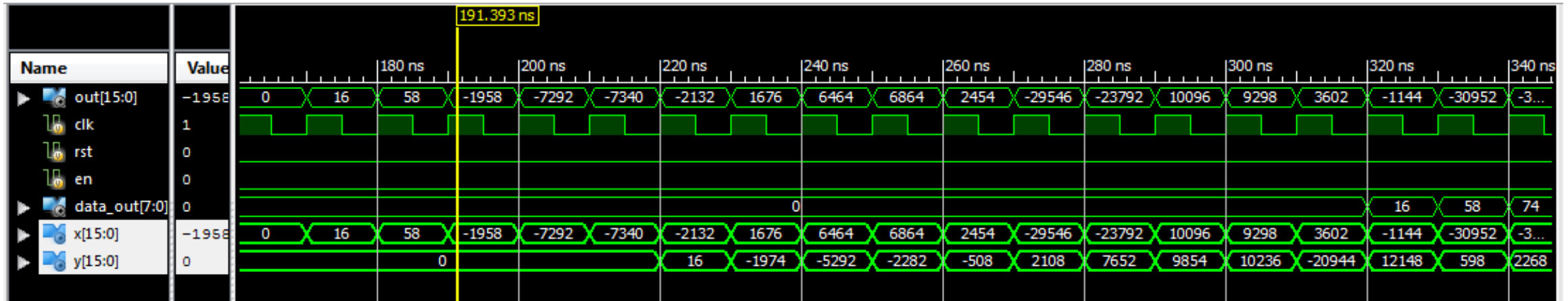
- 4 Tap Filter for Pipelined ICDM



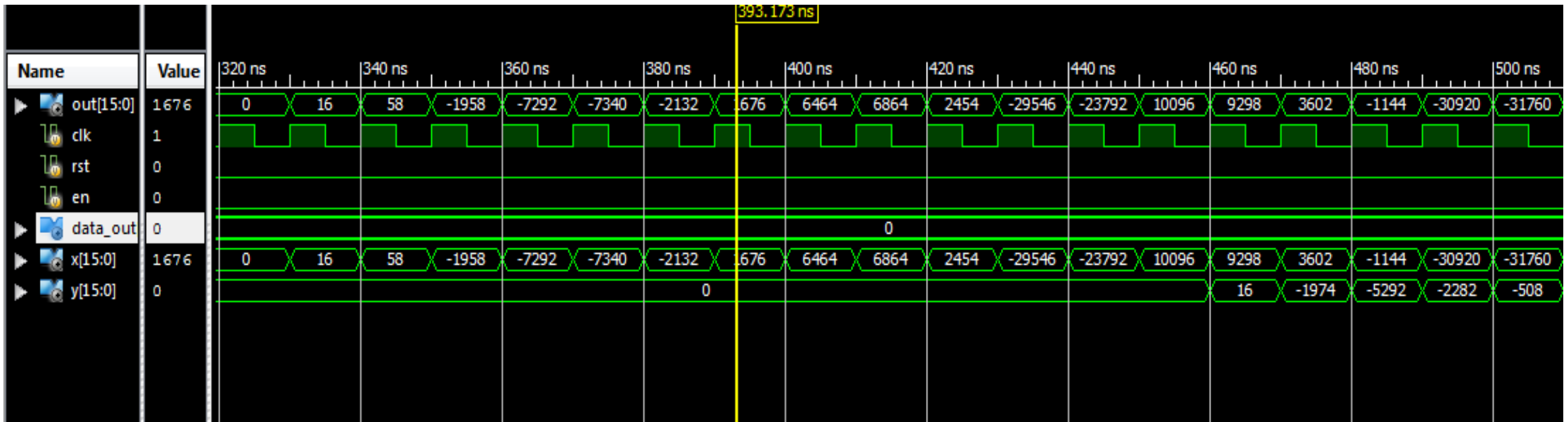
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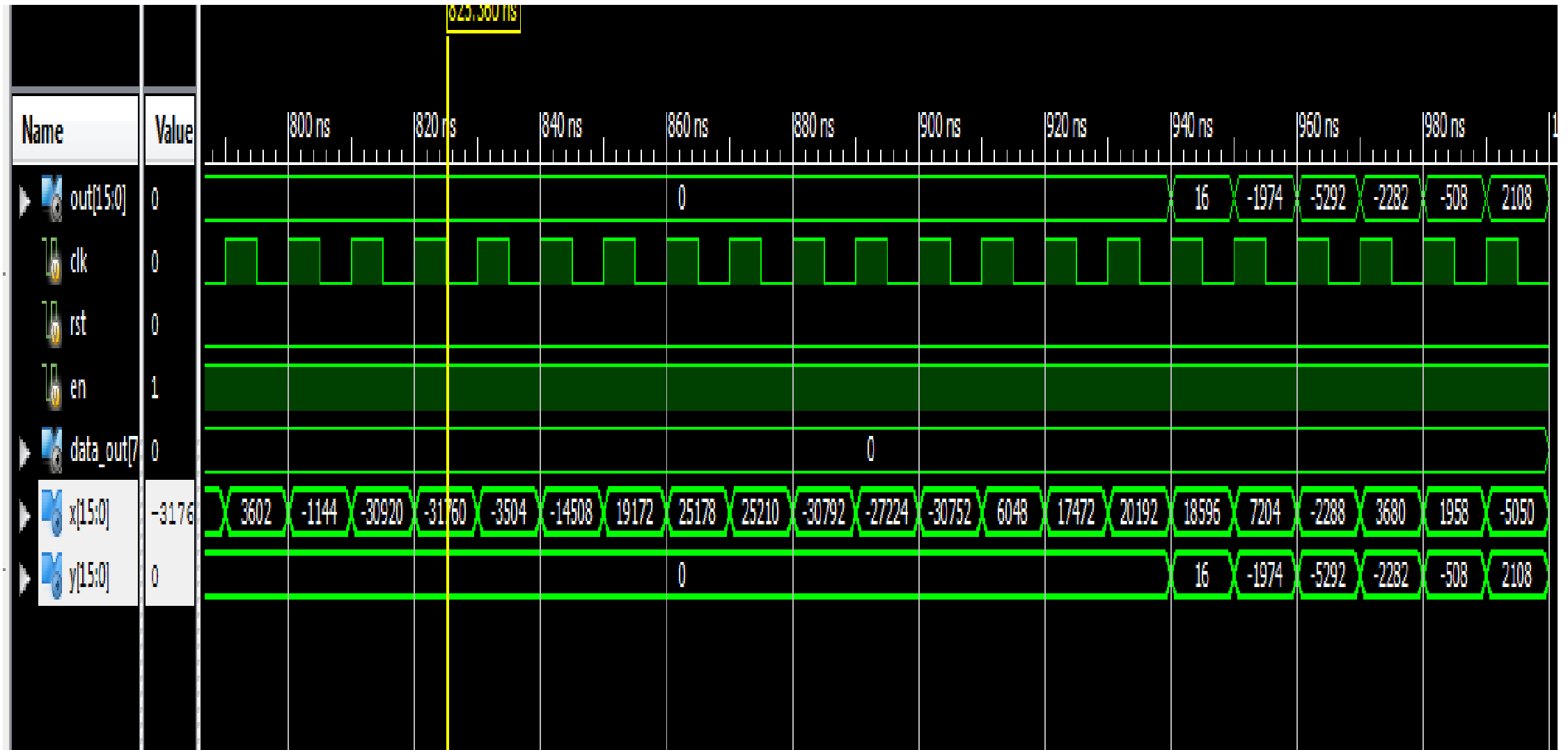
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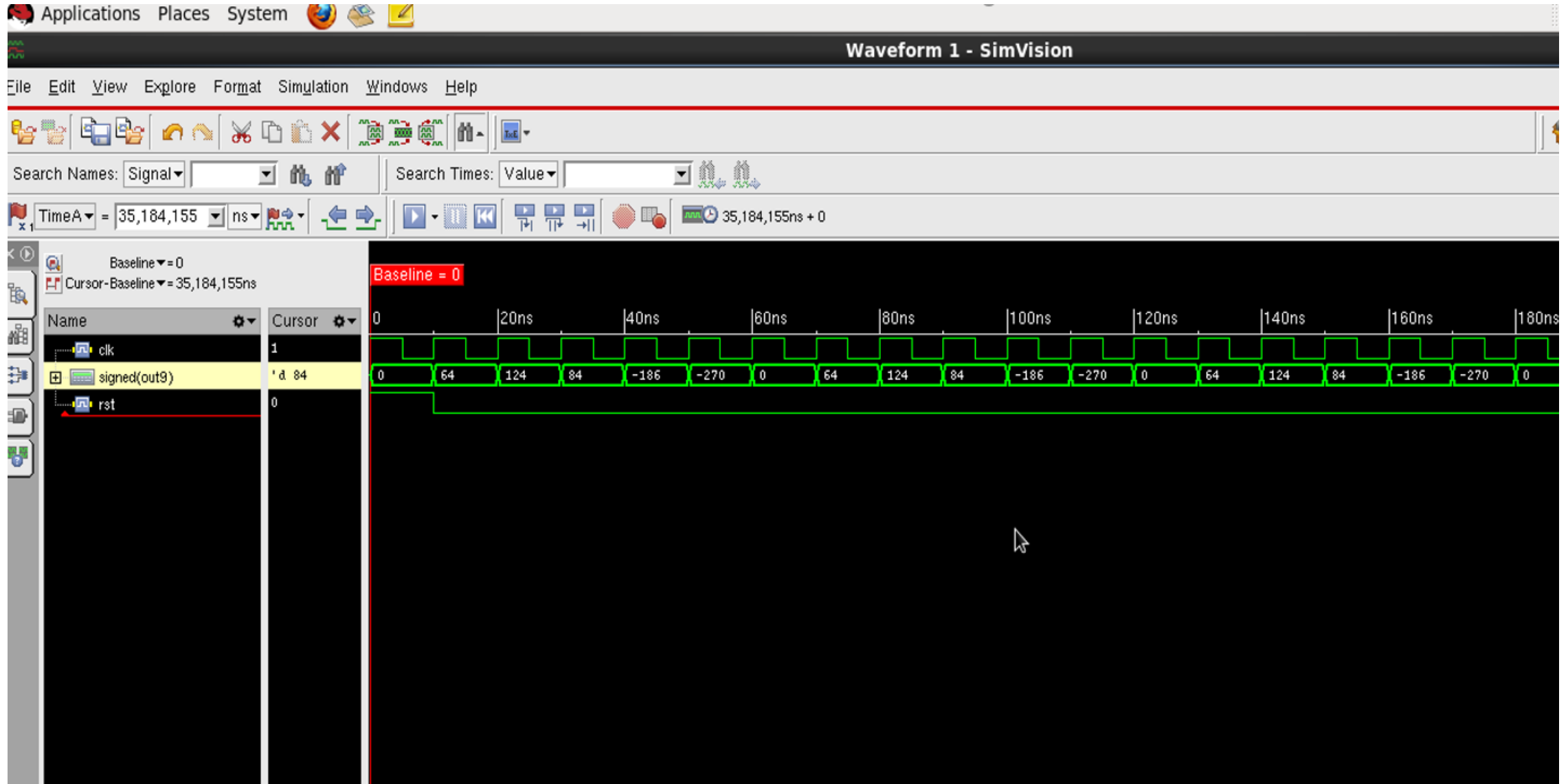
- 32 Tap Filter for Pipelined ICDM



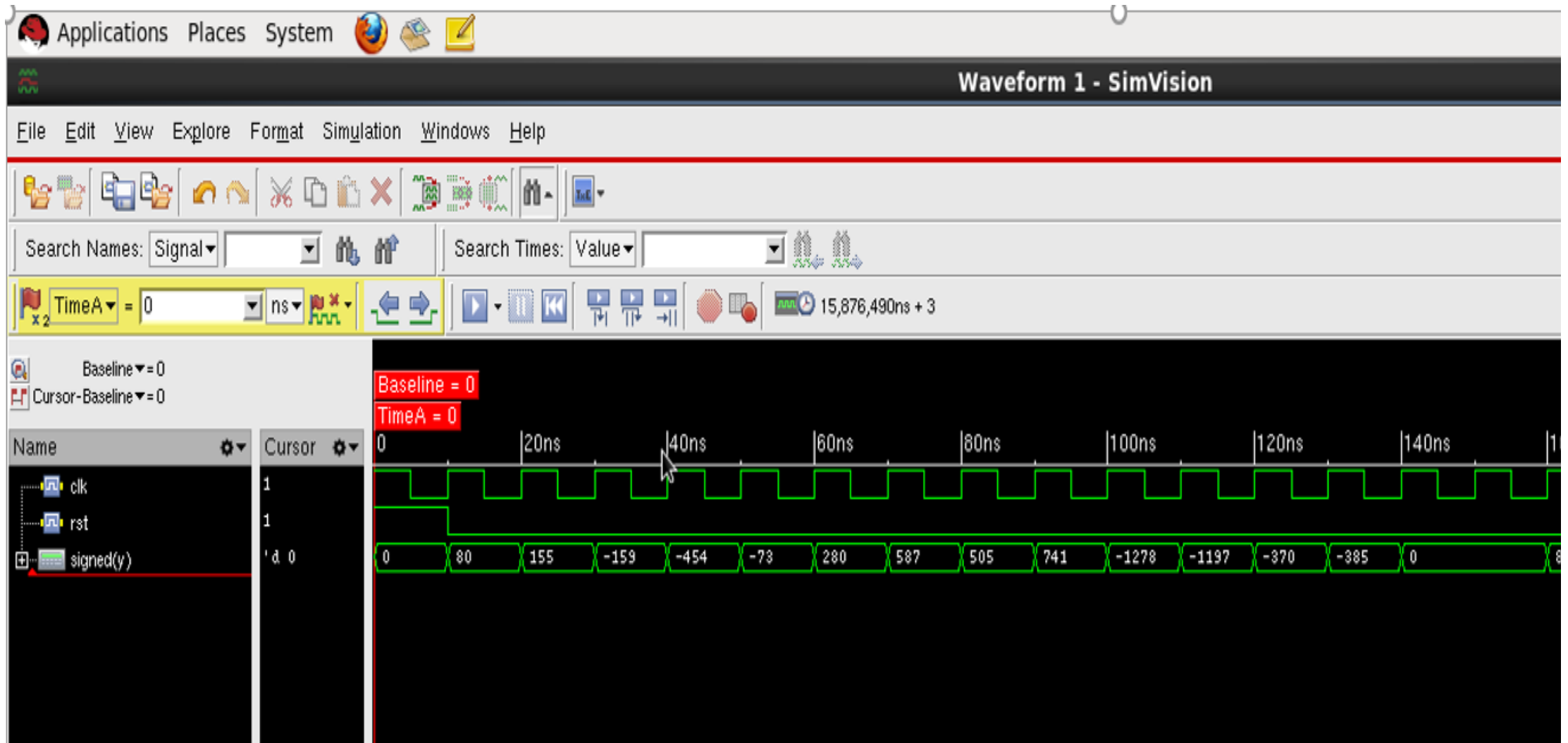
- 64 Tap Filter for Pipelined ICDM



- 4 Tap Filter for ICDM using Cadence



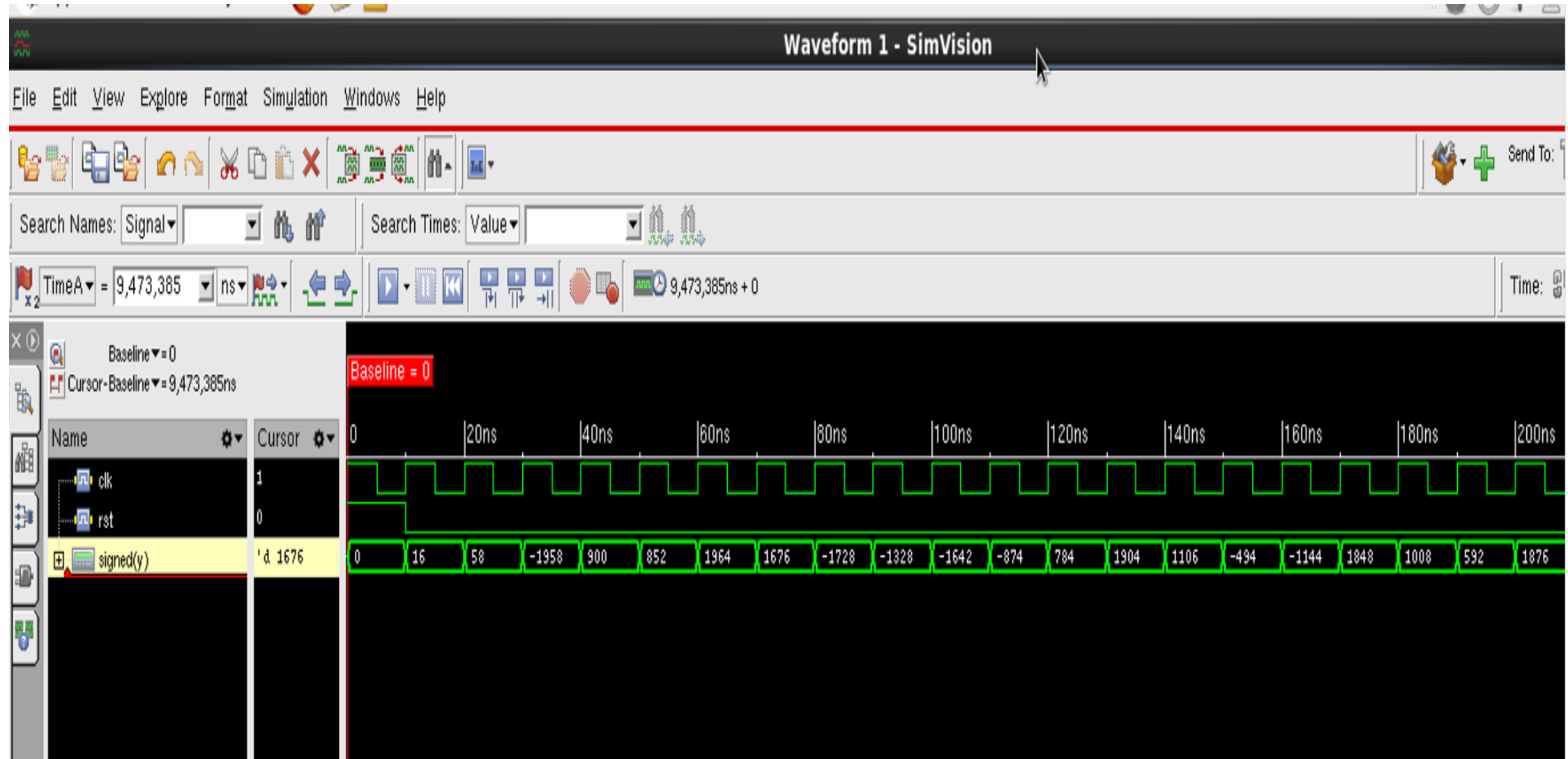
- 8 Tap Filter for ICDM using Cadence



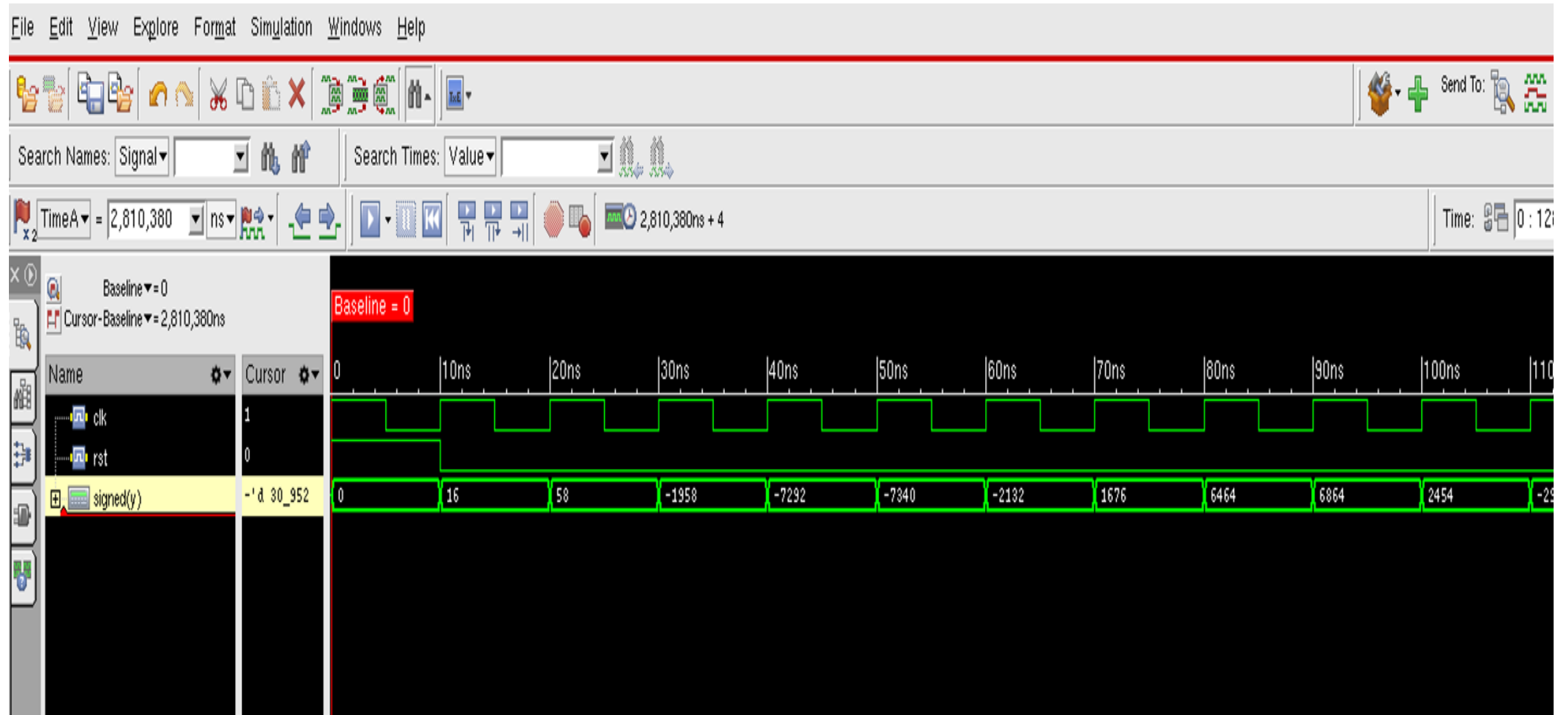
- 16 Tap Filter for ICDM using Cadence



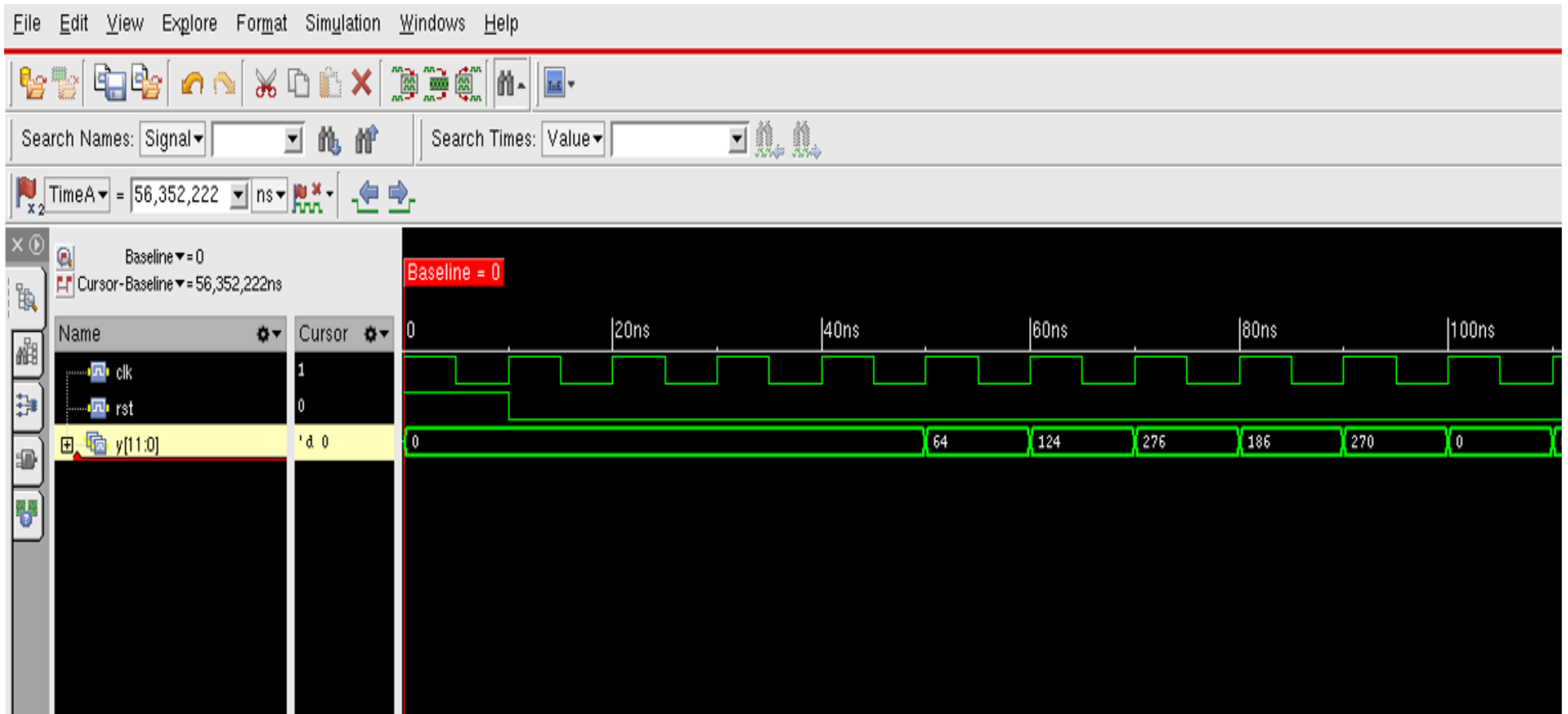
- 32 Tap Filter for ICDM using Cadence



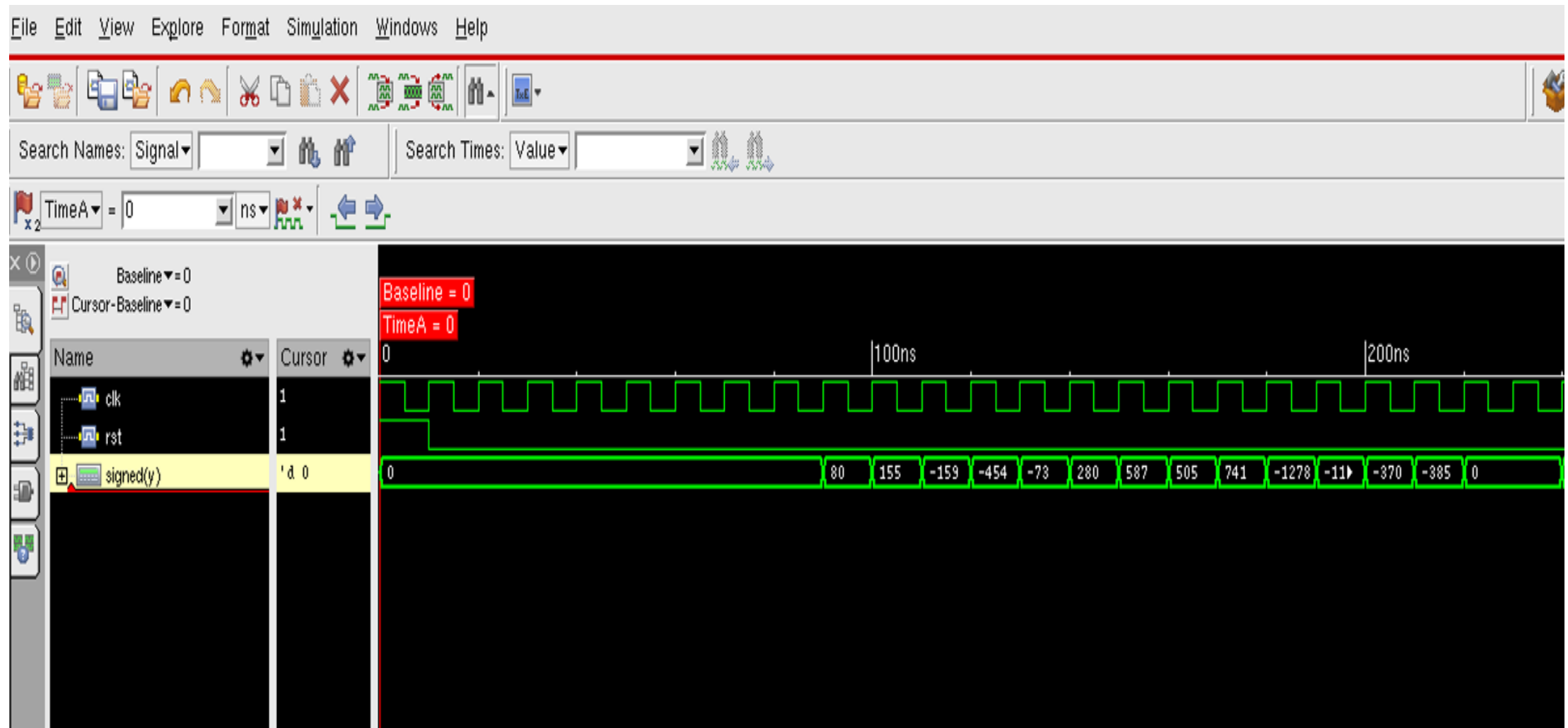
- 64 Tap Filter for ICDM using Cadence



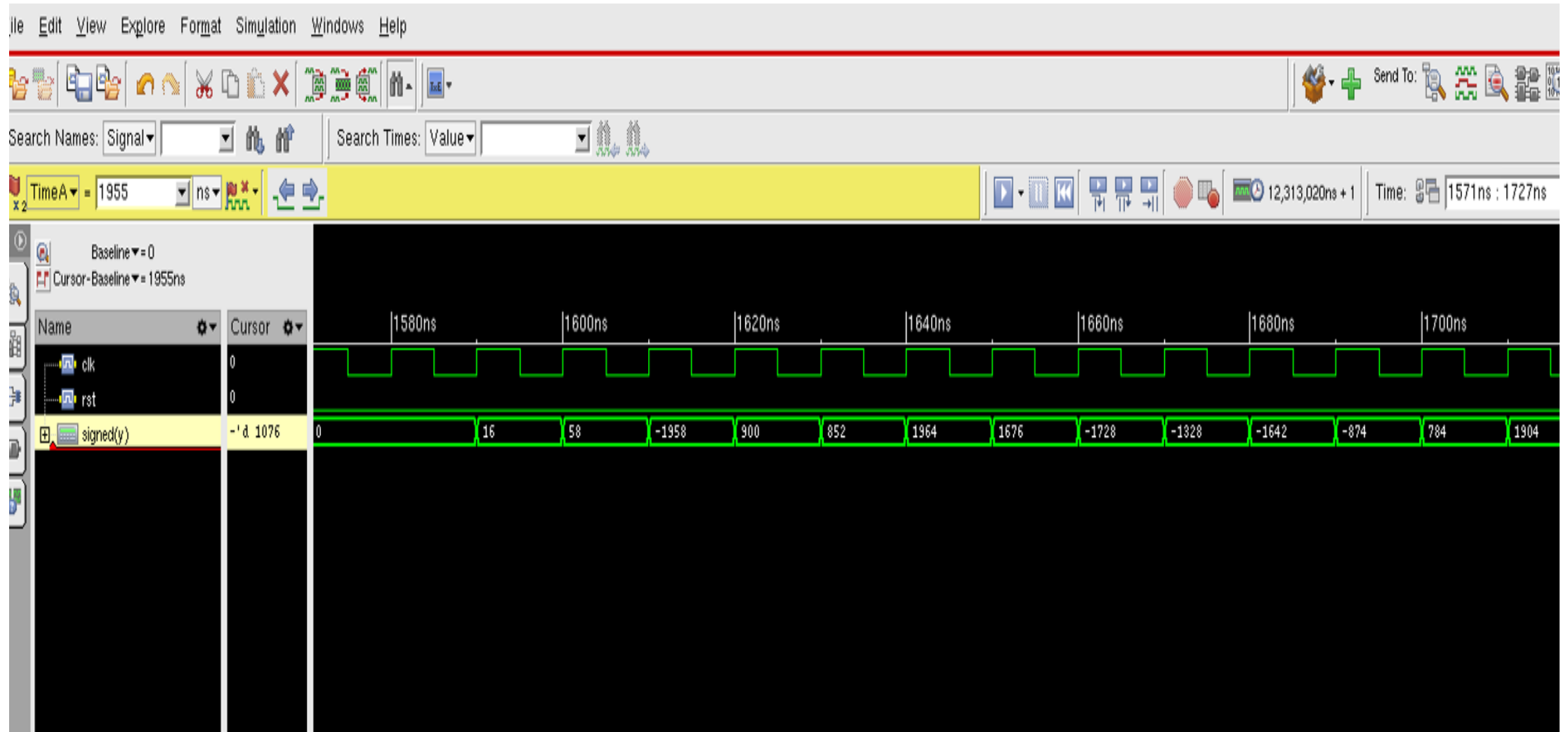
- 4 Tap Filter for Pipelined ICDM using Cadence



- 8 Tap Filter for Pipelined ICDM using Cadence



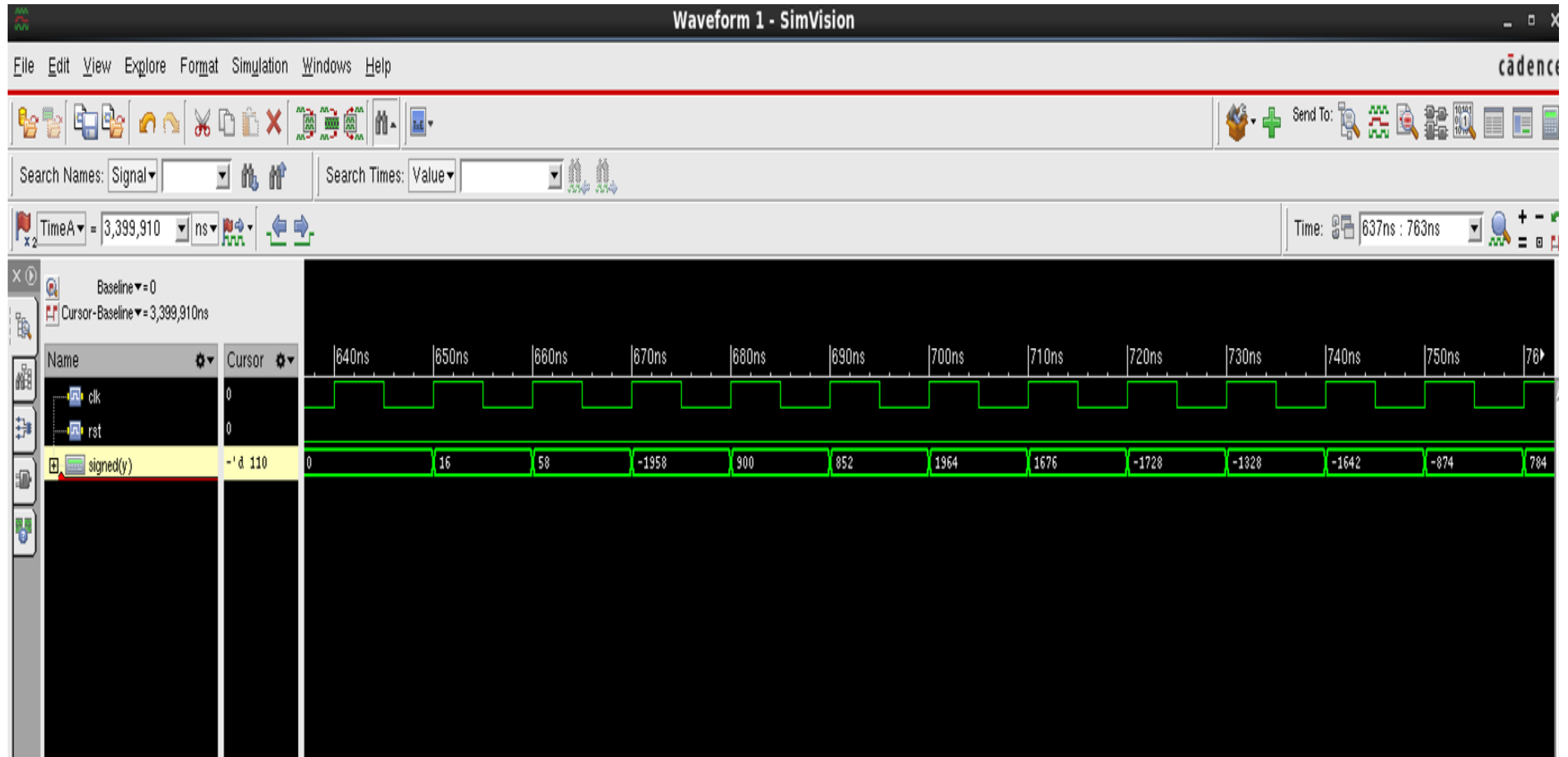
- 16 Tap Filter for Pipelined ICDM using Cadence



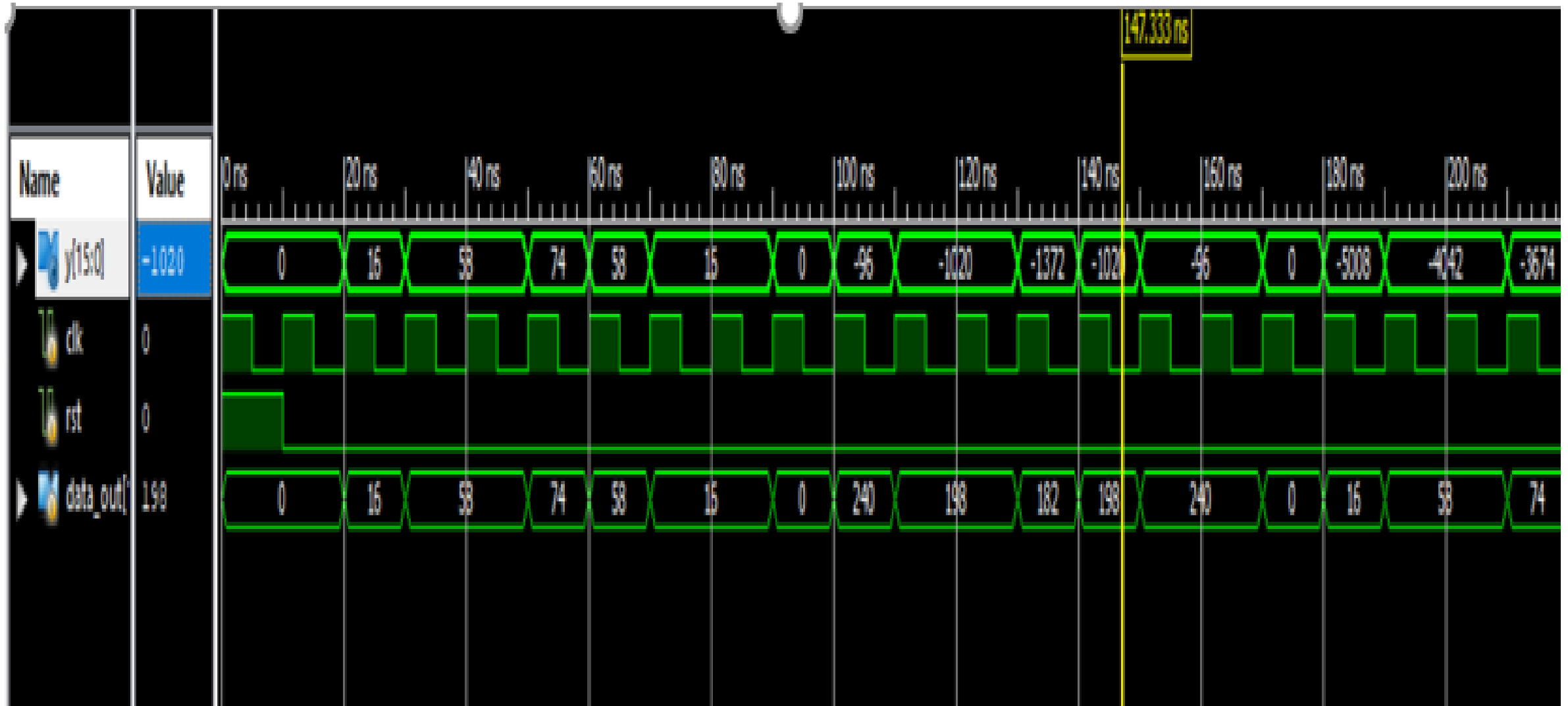
- 32 Tap Filter for ICDM using Cadence



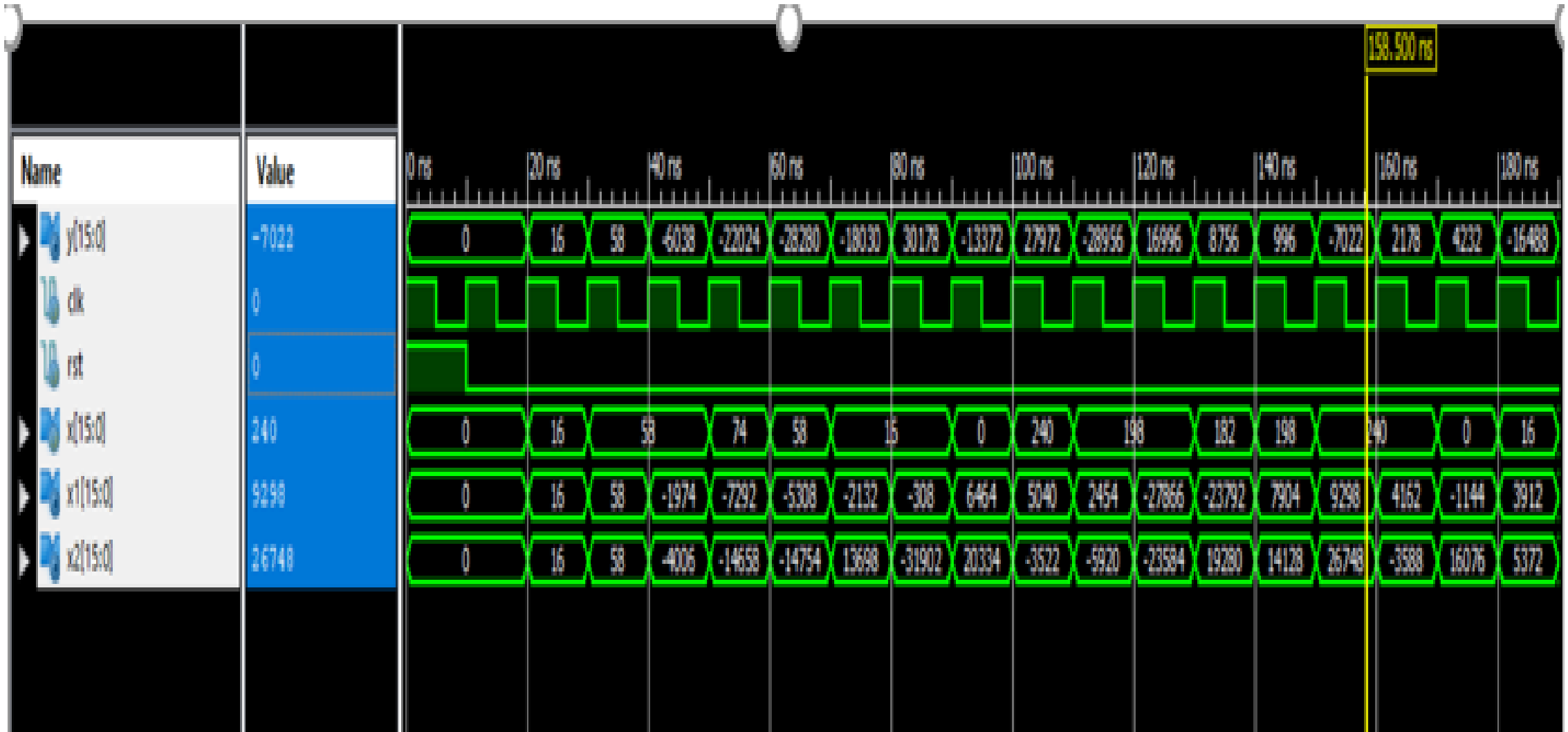
- 64 Tap Filter for Pipelined ICDM using Cadence



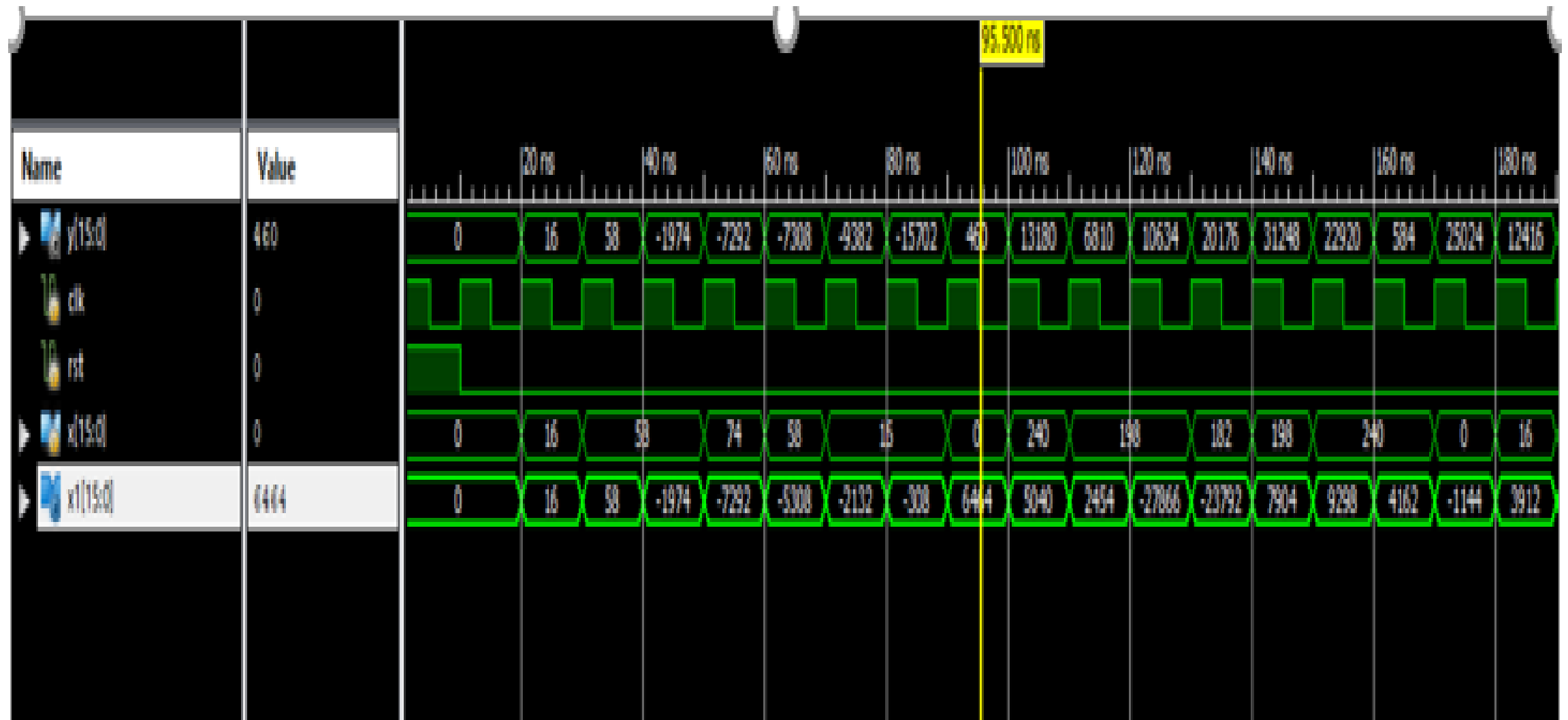
- IEEE 802.16 Decimator factor $M = 1/8$



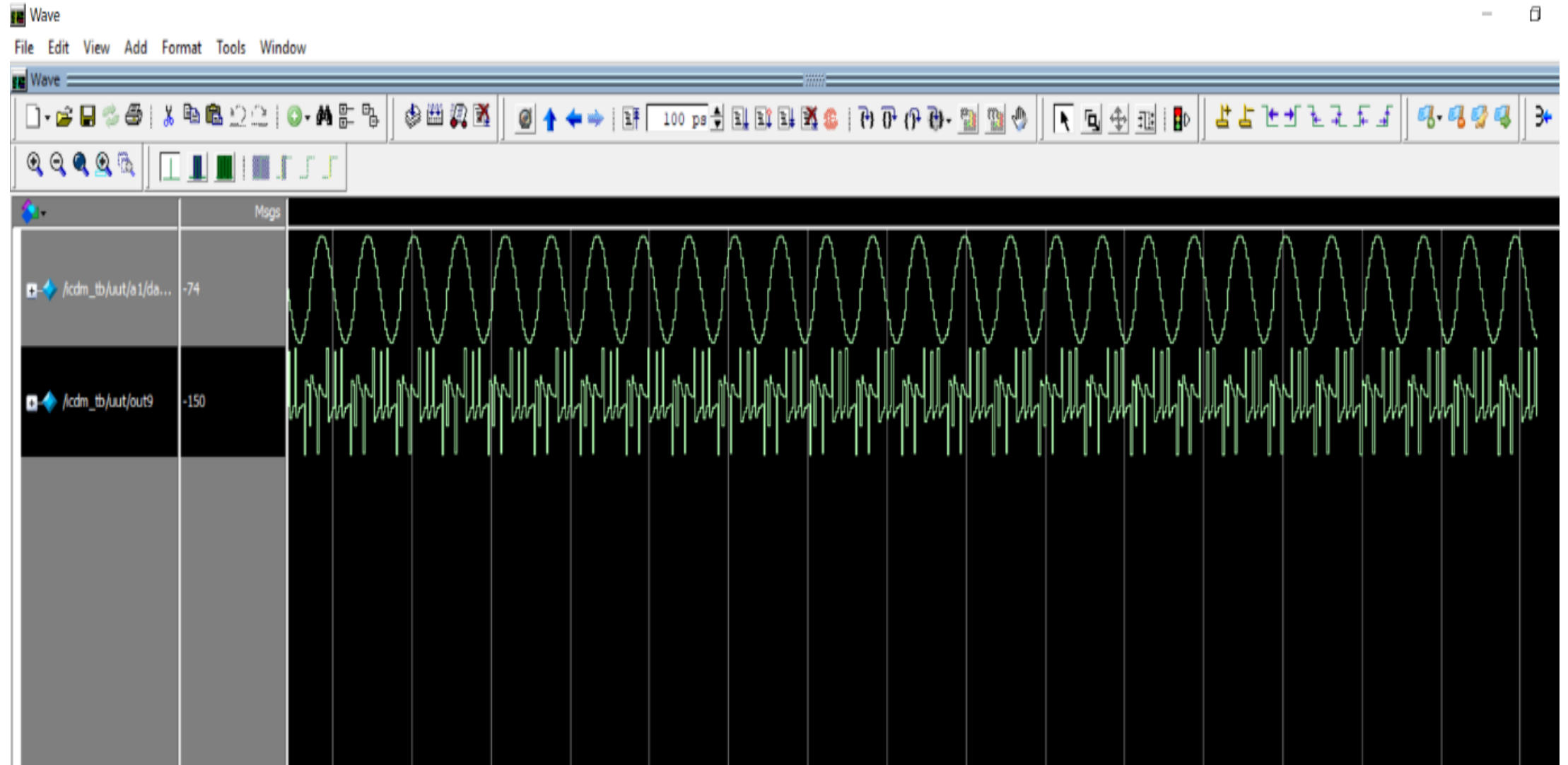
- IEEE 802.16 Decimator factor $M = 1/2 * 1/2 * 1/2$



- IEEE 802.16 Decimator factor $M = (1/4 * 1/2)$



Sine wave Generator using Model sim



Performance comparison using Xilinx

Filter	Parameter	4-Tap	8-Tap	16-Tap	32-Tap	64-Tap
ICDM I, II	Area	72	147	357	598	1141
	Power(W)	57.73m	67.68m	73.04m	83.95m	109m
	Time(s)	4.451n	4.542n	5.038n	5.047n	6.055n
Pipelined I, II	Area	90	271	573	995	2197
	Power(W)	57.93m	78.5m	73.84m	70.41m	80m
	Time(s)	3.662n	4.41n	4.542n	4.874n	5.12n
ICDM	Area	100	234	461	822	1755
	Power(W)	60.14m	75.64m	75.86m	95.54m	152.5m
	Time(s)	3.701n	0.738n	0.907n	1.178n	5.540n
Pipelined	Area	100	234	872	1806	5448
	Power(W)	60.14m	81.63m	77.78m	95.08m	183.44m
	Time(s)	3.701n	0.789n	0.927n	1.102n	4.451n

Performance Comparison using Cadence

Filter	Parameter	4-Tap	8-Tap	16-Tap	32-Tap	64-Tap
ICDM	Area	817	2959	6739	11888	22117
	Power(W)	42942n	0.166m	0.356m	0.777m	1.44m
	Time(s)	2.714n	3.912n	3.912n	3.912n	3.912n
Pipelined	Area	896	4326	8679	17660	35623
	Power(W)	48853n	0.251m	0.4744m	0.9636m	1.938m
	Time(s)	2.714n	3.912n	3.912ns	3.763n	3.763n

TOOLS

- Xilinx ISE 14.7 Design tool
- Cadence

Conclusion

- This project proposes design of the Pipelined ICDM. The coefficient is obtained by FDA tool on the MATLAB. The input sine signal is stored by LUTs. ICDM consists of Mux I and Mux II, add/sub block and decimation selector. Mux I based on ICDM I operation and Mux II based on ICDM II operation.
- Add/sub blocks are used to perform sign reversal of appropriate coefficient in the MCDM. The decimation selector block is used to provide select signal of Mux I and Mux II. Pipelined ICDM architecture reduces time consumption.
- The propagation delay of 64-Tap Pipelining based ICDM filter reducing about 19% when examining with 64-Tap ICDM filter.
- One of the Communication Standard IEEE 802.16 require $1/8$ decimation factor. The decimation result of this Communication Standard is also discussed in this work

References

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Thank you